Cancer Biology

RBM7 deficiency promotes breast cancer metastasis by coordinating MFGE8 splicing switch and NF-kB pathway

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint version 1
February 23, 2024 (this version)

Posted to preprint server
January 3, 2024

Sent for peer review
December 19, 2023

Fang Huang, Zhenwei Dai, Jinmiao Yu, Kainan Wang, Chaoqun Chen, Dan Chen, Jinrui Zhang, Jinyao Zhao, Wenjing Zhang, Yangfan Qi, Yang Wang

Sino-US Research Center for Cancer Translational Medicine of the Second Affiliated Hospital of Dalian Medical University & Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116023, China • Department of Oncology & Sino-US Research Center for Cancer Translational Medicine, the Second Affiliated Hospital, Dalian Medical University, Dalian 116023, China • Department of Pathology, the First Affiliated Hospital of Dalian Medical University, Dalian Medical University, Dalian 116011, China • Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116044, China

Abstract

Aberrant alternative splicing is well-known to be closely associated with tumorigenesis of various cancers. However, the intricate mechanisms underlying breast cancer metastasis driven by deregulated splicing events remain largely unexplored. Here, we unveiled RBM7 as a novel regulator of alternative splicing that plays a crucial role in counteracting the metastatic potential of breast cancer. Through bioinformatics analysis and IHC staining validation, we revealed that RBM7 is decreased in lymph node and distant organ metastases of breast cancer as compared to primary lesions. Furthermore, we found that low expression of RBM7 is correlated with the reduced disease-free survival of breast cancer patients. Breast cancer cells with RBM7 depletion exhibited an increased potential for lung metastasis compared to scramble control cells. The absence of RBM7 stimulated breast cancer cell migration, invasion, and angiogenesis. Mechanistically, RBM7 regulates the splicing of MFGE8 directly, favoring the production of the predominant MFGE8-L isoform. This results in the attenuation of STAT1 phosphorylation and alterations in cell adhesion molecules. The MFGE8-L isoform exerted an inhibitory effect on the migratory and invasive capability of breast cancer cells, while the MFGE8-S isoform had the opposite effect. Particularly, the ectopic expression of MFGE8-L significantly reversed the pro-invasion effect of RBM7 silencing, but did not contribute to the promotion of angiogenesis-related secreted proteins. In RBM7 depleted cells, a gene set enrichment analysis revealed a significant amplification of the NF-κB cascade. Concordantly, RBM7 negatively regulates p65 phosphorylation. Furthermore, an NF-κB inhibitor could obstruct the increase in HUVEC tube formation caused by RBM7 silencing. Clinically, we noticed a positive correlation between RBM7 expression, MFGE-8 exon7 inclusion, and p65 downstream targets. Therefore, our study not only offer mechanistic insights into how abnormal splicing contributes to the aggressiveness of breast cancer, but also provide a new approach for molecular-targeted therapy in combating breast cancer.

eLife assessment

This study presents an **important** finding on the splicing regulatory function of RBM7 and its functional impact in breast cancer metastasis. The evidence supporting the claims of the authors is **solid**, although the inclusion of more delineation of how RBM7 regulates NF-kB and coordinates splicing would have strengthened the study. The work will be of interest to scientists working on breast cancer.

Introduction

Alternative splicing (AS) emerges as a substantial contributor to the production of a myriad of transcripts from limited genes, which enriched protein variation and phenotypic diversity in multicellular organisms. Pre-mRNA splicing reactions require a step-wise assembly of spliceosome, a megadalton complex that recognize consensus regulatory sequences, including 5' and 3' splice sites marking exon-intron boundary, and the branch point site (BPS), to accomplish exon definition (Shi, 2017) (Wilkinson, Charenton et al., 2020). While a splice site is suboptimal, AS emerges under control of inherent sequence acting in cis, namely exonic or intronic splicing enhancer or silence, and the cognate trans-acting splicing factors, which can strengthen or weaken the recognition of splice sites by spliceosome.

It is well documented that AS dysregulation leads to multiple human diseases, including cancer (Montes, Sanford et al., 2019) (Singh & Cooper, 2012). Tumorigenicity has been linked with cancer-specific splicing switch arising from either cis-elements mutations or deregulated splicing regulatory factors. In some cases, splicing factors, particularly spliceosomal components SF3B1 and U2AF1, are frequently mutated across diverse cancer types (Baralle, 2005) (Harbour, Roberson et al., 2013) (Smith, Choudhary et al., 2019) (Cheruiyot, Li et al., 2021). For instance, recurrent SF3B1 mutations led to retention of intronic region in BRD9 pre-mRNA, conferring loss of core subunit function to the ncBAF chromatin remodeling complex and metastatic advantage to melanoma cells (Inoue, Chew et al., 2019). Another layer of regulation involves the abnormal expression of splicing factors derived from altered regulatory pathway. As an archetypal example, a set of members in SR protein family such as SRSF1 and SRSF3 are upregulated in breast tumors, lung tumors, glioma and exhibit oncogenic ability via altering AS products (for example, Bcl-x, MYO1B, NUMB and p53) (Anczuków, Akerman et al., 2015) (Zhou, Wang et al., 2019) (Tang, Horikawa et al., 2012) (Zhou, Gong et al., 2020). Otherwise, many splicing factors such as RBM proteins can either functions as oncoproteins or tumor suppressors in cancer-type dependent manner (Li, Guo et al., 2021), which may be attributed to tissue-specific AS patterns.

Breast cancer is the most common diagnosed malignancy and remain the second leading cause of cancer mortality (over 41,000 deaths annually in the USA) among females worldwide (Siegel, Miller et al., 2019). Metastasis to vital organs accounts for approximately 90% cancer-related death. The 5-year overall survival rate of patients with metastatic breast cancer is drastically decreased to about 25%, as compared to the primary cancer patients' prognosis with a survival rate greater than 80% (Rabbani & Mazar, 2007) (Valastyan & Weinberg, 2011). Despite advances in early diagnosis and comprehensive interventions for breast cancer treatment, limited effective strategy against metastasis has been an outstanding challenge for improving prognostication of patients. This highlights the urgent need for deepened understanding of mechanisms underlying development of metastasis. Tumor metastasis is multi-step process involving a sequential cascade of events, such as epithelial-mesenchymal transition (EMT), local invasion from primary tumor site, lymphogenic or hematogenic dissemination, cancer cells seeding and metastatic lesions formation (Liang, Zhang et al., 2020). AS dysregulation has been implicated in multistage process of metastasis. For example, the switch of CD44 or FGFR2 splicing isoforms can lead to EMT phenotype (Chen, Zhao et al., 2018) (Warzecha, Sato et al., 2009). VEGF-A or XBP1 splicing transition is tightly linked to angiogenesis in cancer (Bates, 2008) (Zeng, Xiao et al., 2013). On account of the phenotypic plasticity in metastatic cells conferred by alternative splicing, further investigation in potential role of critical splicing regulatory factors and the associated splicing events in breast cancer metastasis is warranted.

RNA-binding proteins (RBPs) are a large class of conserved proteins that participate in cancer development through precisely control RNA metabolism, including RNA splicing. RBM7, a member of RBP family, is commonly known to exert RNA surveillance activity as a component of the human nuclear exosome target (NEXT) complex (Meola, Domanski et al., 2016). Aberrant expression of RBM7 is not only associated with lung fibrosis onset and motor neuronopathy, but also impacts cancer cell proliferation (Fukushima, Satoh et al., 2020) (Xi, Zhang et al., 2020). Previous findings have established a direct connection of RBM7 with spliceosome factors SF3B2 and SF3B4 (Falk, Finogenova et al., 2016), yet whether RBM7 is involved in the regulation of AS remain elusive. In this study, we report that RBM7 is frequently reduced in a subset of breast cancer metastases from lymph nodes lesions and distant organ as compared to primary tumors and its low expression predicts dismal survival outcomes of patients. Breast cancer cells with loss of RBM7 gain enhanced aggressive capability and metastatic potential. Mechanistically, RBM7 ablation enhanced the splicing switch of MFG-E8 pre-mRNA from canonical isoform MFGE8-L to a truncated isoform MFGE8-S. Ectopic expression of MFGE8-L caused marked suppression of migratory and invasive ability of breast cancer cells, whereas MFGE8-S showed opposite function/ disabled such function. At another layer, RBM7 discordantly regulated the phosphorylation of p65. In concert, NF-κB inhibitor abolished pro-angiogenesis effect of RBM7 ablation in breast cancer. Above all, our findings uncovered a previously unidentified role of RBM7 in breast cancer metastasis via coupling MFGE8 splicing switch and NF-κB signaling activation, suggesting a new therapeutic target and prognostic predictor for breast cancer.

Results

RBM7 is reduced in metastatic lesions of breast cancer and positively correlated with patients' prognosis

To determine the clinical relevance of RBM7 in breast cancer, we assessed the correlation between RBM7 and important clinical outcomes. Analysis of TCGA, GEO and the METABRIC breast cancer datasets showed that RBM7 expression positively correlates with overall survival of breast cancer patients (**Figure 1A** and **Supplementary Figure 1A-1B**). RBM7 low expression predicts the reduced disease-free survival in patients with breast cancer (**Figure 1B** and **Supplementary Figure 1C**). Congruently, we found RBM7 is decreased in breast cancer compared to normal tissues (**Figure 1C** and **Supplementary Figure 1D**). Further dissection in a cohort of breast cancer patients from the cBioPortal unveiled cases with lymph node metastasis have lower expression of RBM7 as compared to cases without metastasis (**Figure 1D**). We next performed immunohistochemistry (IHC) detection of RBM7 in breast cancer tissue microarrays. The results showed that patients bearing clinical stage III tumors have significantly decreased RBM7 expression than patients bearing I and II clinical stage tumors (**Figure 1E**). In tissue chip containing triple negative breast cancer, the more aggressive and mesenchymal breast cancer subtype, RBM7 exhibits reduced protein levels in tumors as compared to paracancer tissues (**Figure 1F**). In addition, we collect specimen consisted of tumor tissues from 21 breast primary cancer, in 9 cases, from metastatic loci and took advantage of IHC analysis to determine the RBM7 expression. Most (25% strong positive;67% moderate positive) of primary tumor samples displayed a positive IHC staining for RBM7, whereas most of lung/liver metastatic lesions showed weak or negative RBM7 expression (**Figure 1G**). We also obtained paired primary breast cancer and lymph node metastatic loci and found, when compared to the corresponding primary tumor, the expression of RBM7 is significantly decreased in lymph node metastases (**Figure 1H**-**1I**). These results associated RBM7 with breast cancer metastasis, prompting us to further investigate the functional role of RBM7 in metastatic processes of breast cancer.

RBM7 silencing provokes metastatic potential of breast cancer

In light with the above findings, we generated RBM7-silenced breast cancer cells using lentiviral expression of short hairpin RNA (shRNA) followed by profiling high-throughput transcriptome by RNA sequencing. Strikingly, Gene Ontology and pathway analysis displayed a wide range of RBM7-regulated genes were enriched in extracellular matrix organization, cell migration and angiogenesis (**Figure 2A**-**2B**), which have been characterized as crucial processes in tumor metastasis. A heatmap generated from qRT-PCR analysis of differentially expressed genes in RBM7-depleted breast cancer cells verified that a series of metastasis-related genes were upregulated upon RBM7 knockdown (**Figure 2C**). The lung metastasis is one of the most prevalent distant metastases that breast cancer is prone to and associated with poor prognosis of patients (Medeiros & Allan, 2019). Thus, we performed tail-vein transfer model to evaluate the potential impact of RBM7 on pulmonary metastasis of breast cancer cells. 5×10^4 4T1 cells, the mouse breast cancer cell line simulating stage IV human breast tumor, with stable RBM7 silence or control vector were implanted into BALB/c mice via tail vein. After 3 weeks, RBM7-silenced 4T1 cells were able to develop more spontaneous lung metastases than control cells as judged by pulmonary nodules number and haematoxylin-eosin (HE) staining of lung tissues (**Figure 2D**-**2F** and **Supplementary 2A**). Cancer metastasis is comprised of a complex cascade of events, such as migration, invasion and angiogenesis, thus we next to determine which process is mediated by RBM7-loss in facilitating breast cancer metastasis. We utilized transwell chamber with or without matrix coat and would healing assays to examine the potential function of RBM7 in migration and invasion capabilities and found the depletion of RBM7 led to a visible increase in mobility and invasiveness of both the triple-negative breast cancer (TNBC) cell line MDA-MB-231, estrogen receptor positive breast cancer cell line MCF7 and 4T1 cells (**Figure 2G**-**2H** and **Supplementary 2B**-**2E**). Conversely, RBM7 overexpression notably repressed MDA-MB-231 and MCF7 cells to migrate or invade across transwell inserts (**Figure 2I**-**2J**). Similar results were obtained in two additional TNBC cell lines BT549 and HCC1937 (Supplementary 2F-2G), which proved that RBM7 possessed the capability to suppress the migration and invasion of breast cancer cells. To examine the impact of RBM7 on angiogenesis in breast cancer, we cultured human umbilical vein endothelia cells (HUVEC) with condition medium obtained from culture supernatants of RBM7-depleted or scramble shRNA transduced cells for tube formation analysis. Compared to the scramble control, HUVEC incubated with culture medium from RBM7-knockdown breast cancer cells exhibited accelerated tube formation length and branch junction (**Figure 2K**), suggesting the inhibitory effect of RBM7 on breast cancer angiogenesis. Taken together, the above data demonstrated RBM7 as tumor suppressor by regulating breast cancer migration, invasion and angiogenesis.

RBM7 exhibits splicing regulatory function in breast cancer

In previous studies, RBM7 is commonly characterized as an RNA-binding constituent of trimeric NEXT complex, which directs the RNA exosome to non-coding transcriptome for degradation (Lubas, Andersen et al., 2015). However, the mechanistical role of RBM7 in cancer remain largely unknown. Intriguingly, we observed that, attributed to the ablation of RBM7, almost 551 splicing events occurred significant decrease in PSI (percent splicing inclusion), whereas 289 splicing events endured increased PSI (**Figure 3A**). The changed events were divided among various types of AS (alternative splicing), including SE (skipping exon), IR (intron retention), A5SS (alternative 5' splice site), A3SS (alternative 3' splice site), and MXE (mutually exclusive exons), of note, SE events were the most prominently regulated (**Figure 3B**). A majority of AS events in particular of SE and A5SS category were in decrease of PSI upon RBM7 depletion (**Figure**

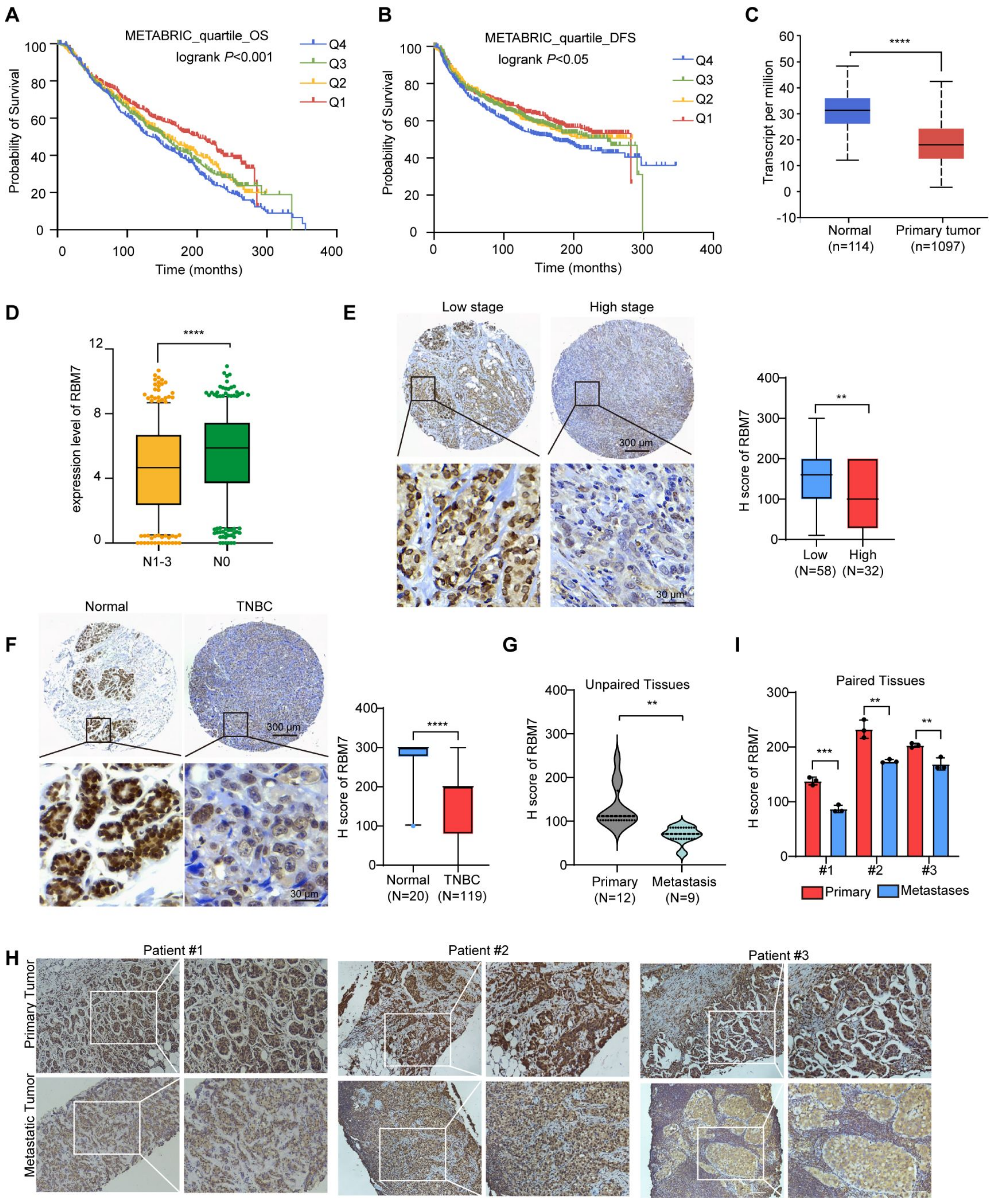


Figure 1.

Decreased RBM7 expression was associated with poor prognosis of breast cancer.

(A-B) The correlation between RBM7 mRNA expression and overall survival (OS) and disease-free-survival (DFS) of breast cancer patients (n=1980) based on the METABRIC dataset. The samples were divided into four equal parts, containing lower quartile Q1, median quartile Q2, upper quartile Q3 and higher quartile Q4 according to the expression of RBM7. (C) RBM7 mRNA level was analyzed in primary breast carcinoma (n=1097) and normal tissues (n=114) from the UALCAN dataset. (D) RBM7 expression level was analyzed in breast invasive carcinoma based on nodal metastasis status from TCGA dataset. BRCA samples were classified into N0 (No regional lymph node metastasis) (n=515), and metastases in axillary lymph node (N1-N3) (n=565). (E) Representative images of IHC staining with RBM7 expression in high (n=32) and low (n=58) clinical stages in breast cancer tissue microarray. Scale bars = 300 μm (top) or 30 μm (bottom). (F) Representative images of IHC staining with RBM7 expression in Triple-Negative Breast Cancer (n=119) and para-carcinoma tissue (n=20) in our tissue microarray. Scale bars = 300 μm (top) and 30 μm (bottom). (G) Quantitative analysis of RBM7 expression by IHC staining in primary breast cancer (n=12) and unpaired metastatic cancer (n=9). (H-I) Representative immunostaining images of RBM7 in 3 paired primary breast cancer and lymphatic metastasis. Scale bars = 100 μm. The p value was obtained by log-rank test (A, B), and unpaired Student's T test (C, D, E, F, G, I). Data are presented as means ± SD.

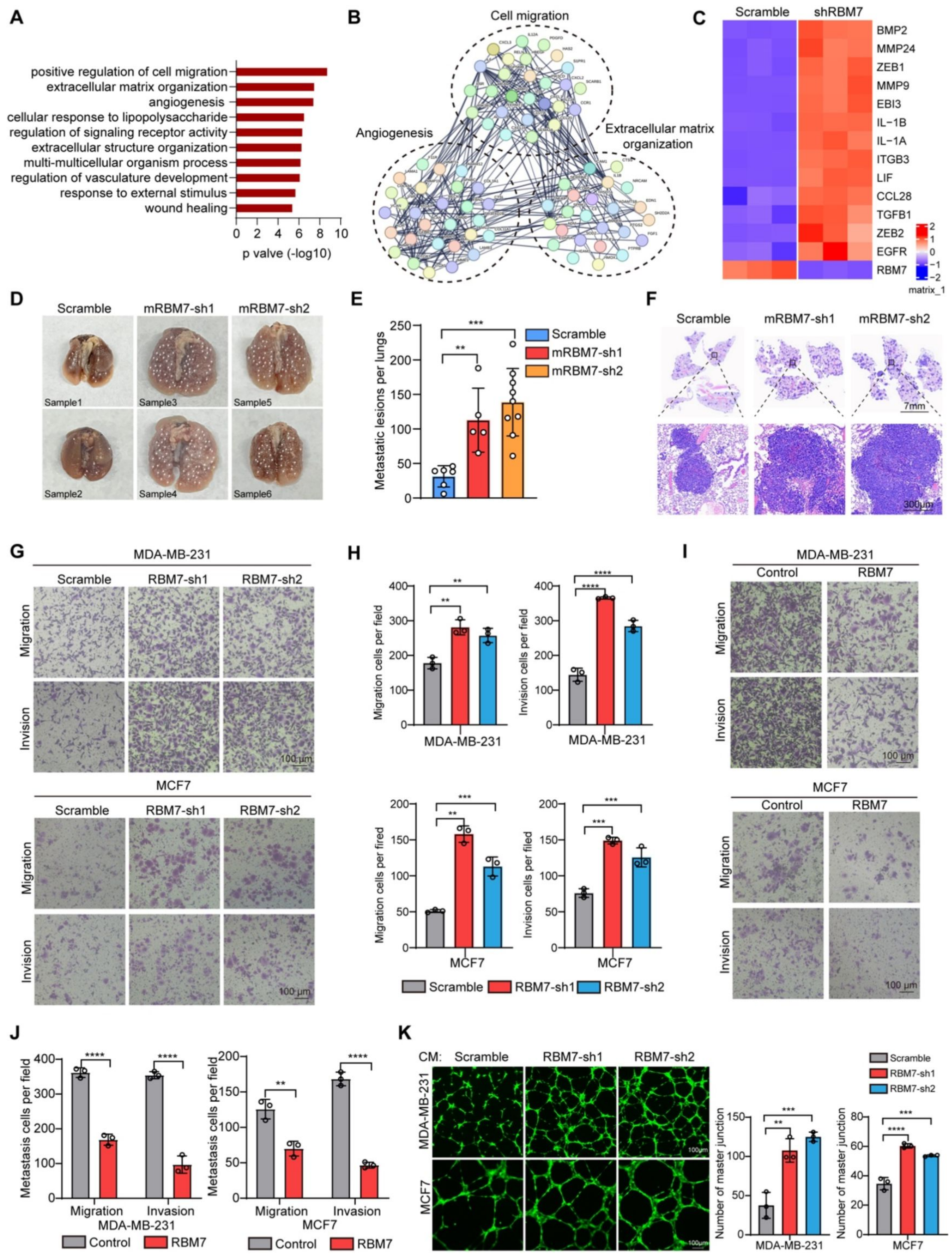


Figure 2.

RBM7 negatively regulated breast cancer metastatic potential.

(A) Gene Ontology analysis showed the significantly affected biological process upon RBM7 knockdown in MDA-MB-231 cells. (B) Functional association network of RBM7-regulated targets. The genes were analyzed using the STRING database, and subgroups are marked according to their function. (C-E) 4T1 cells without or with RBM7 knockdown were injected into tail vein of immunodeficient NYG mice to establish a lung metastasis model. And spontaneous pulmonary metastases were assessed after about 2 weeks. (C)The white arrow indicates lung lesions macroscopically. (D) H&E stained lung sections were quantified for the number of spontaneous metastatic lesions from NYG mice (n=6, 5 and 9). Data are mean ± SD. P values were determined by one-way ANOVA with Dunnett's multiple comparisons test. (E) Representative images of H&E stained lung sections are shown. Scale bars=7 mm (top) and 300 μm (bottom). (F-I) The metastatic ability of RBM7 for breast cancer cells was evaluated by Transwell assay. Data are mean ± SD from three random fields. P values were determined by unpaired Student's T test (I) or one-way ANOVA with Dunnett's multiple comparisons test (G). (J) Representative images of the tube formation of HUVEC cells, treated with conditional medium, which cultured RBM7-KD cells and control cells for 48h and collected. Scale bars: 100 μm. Quantification of master junction was by imageJ software.

3C -**3D**), indicating that the presence of RBM7 may promote the usage of alternative splice sites. Subsequently, we extracted the candidate binding motifs which reside within cassette exons or A5SS and found UUUCUU motfs, simiar to RBM7 binding site (polyU) were enriched in RBM7 positively regulated AS events to a greater extent that control exons unimpacted by RBM7 (**Figure 3E**), raising the possibility that RBM7 may serve as a splicing activator through directly tethering cis-element in pre-mRNA. This is coherent with the nuclear localization of RBM7 in breast cancer cells (Supplementary 3A). In order to explore the splicing regulation activity of RBM7 in distinct pre-mRNA context, we validated a series of RBM7-significantly-changed splicing events across different AS types according to the RNA-seq results. The RT-PCR analysis demonstrated that the PSI of selected gene related to cancer progression was either increased or decreased endogenously upon RBM7 depletion (**Figure 3F** and **Supplementary 3B**). Considering that the AS is a sophisticated multi-step reaction that is regulated by both pre-mRNA-interacting splicing factor and other assistant factors in a context-dependent manner (Wright, Smith et al., 2022), we doubted the divergent effect of RBM7 on distinct splicing targets was due to a more complicated mechanism. Gene ontology (GO) enrichment analysis identified a majority of genes undergoing AS alterations were involved in cytoskeleton assembly and organization (**Figure 3G**). While globally analyzing the network enrichment of RBM7-regulated downstream splicing events, we observed its targets are functionally connected to well-linked network containing gene subgroups (**Figure 3H**), which are associated with biological processes in tumor cell metastasis, again suggesting the cancer-related function of RBM7.

RBM7 depletion switches splicing outcomes of MFG-E8 by decreasing MFG-E8-L isoform

To ascertain the functional targets that explain the inhibitory role of RBM7 in breast cancer metastasis, we examined RNA immunoprecipitation -seq for RBM7 and RNA-seq data for breast cancer cells expressing two shRNAs targeting RBM7. Venn diagram illustrating overlapping genes pointed forward a hotspot gene, namely MFGE8 (**Figure 4A**). We focused on the AS control of MFGE8 by RBM7 resulting in a significant alteration in ratio of MFGE8 two isoforms, of which the canonical isoform (MFGE8-L) has been demonstrated to be correlated with tumor formation and aggressiveness of breast cancer (Yang, Hayashida et al., 2011). Importantly, the exon 7 located in F5/8 type C 2 domain, which is obligated to bind to phosphatidylserine, was lacked in the short isoform of MFGE8 (**Figure 4B**). Further RT-PCR identification verified RBM7 ablation indeed potentiated the production of MFGE8 short splice variant (MFGE8-S) in various TNBC cell lines (**Figure 4C**). Similar results were obtained from MCF7 cells, suggesting that RBM7 regulated the AS of MFGE8 independently of the ER status. In separate experiments, we transfected breast cancer cells with an exogenous MFGE8 splicing reporter containing exon 6-8 and 600bp of flanking intron sequences to determine the splicing regulatory function of RBM7. Accordingly, the depletion of RBM7 still induced the splicing shift of the minigene reporter by elevating MFGE8-S variant (**Figure 4D**). To test whether such splicing regulatory effect was directly or indirectly, we analyzed the sequence of alternative exon 7 and the motifs nearby its 5' or 3' splice sites, and found two poly(U) motifs are positioned in proximal to the pseudo 3' splice site. Subsequent RT-PCR assay for the precipitation in RIP assays confirmed RBM7 could bind to the upstream sequence containing UUUCUU motifs adjacent to 3' splice site of MFGE8 cassette exon, but not another region nearby it (**Figure 4E**). To pinpoint the location for the potential cis-element for AS regulation by RBM7, we designed antisense oligonucleotides (ASOs), which have been chemically modified to avoid endogenous nuclease degradation without recruiting RNase H, to block the UUUCUU residues in proximity to intron6/exon7 junction. As shown in **Figure 4F** , when compared to control ASO, treatment with ASOs targeting the potential binding site (UUUCUU) of RBM7, contributed to the exclusion of exon7, further suggesting that RBM7 regulated the exon7 skipping of MFGE8 by directly binding to its pre-mRNA.

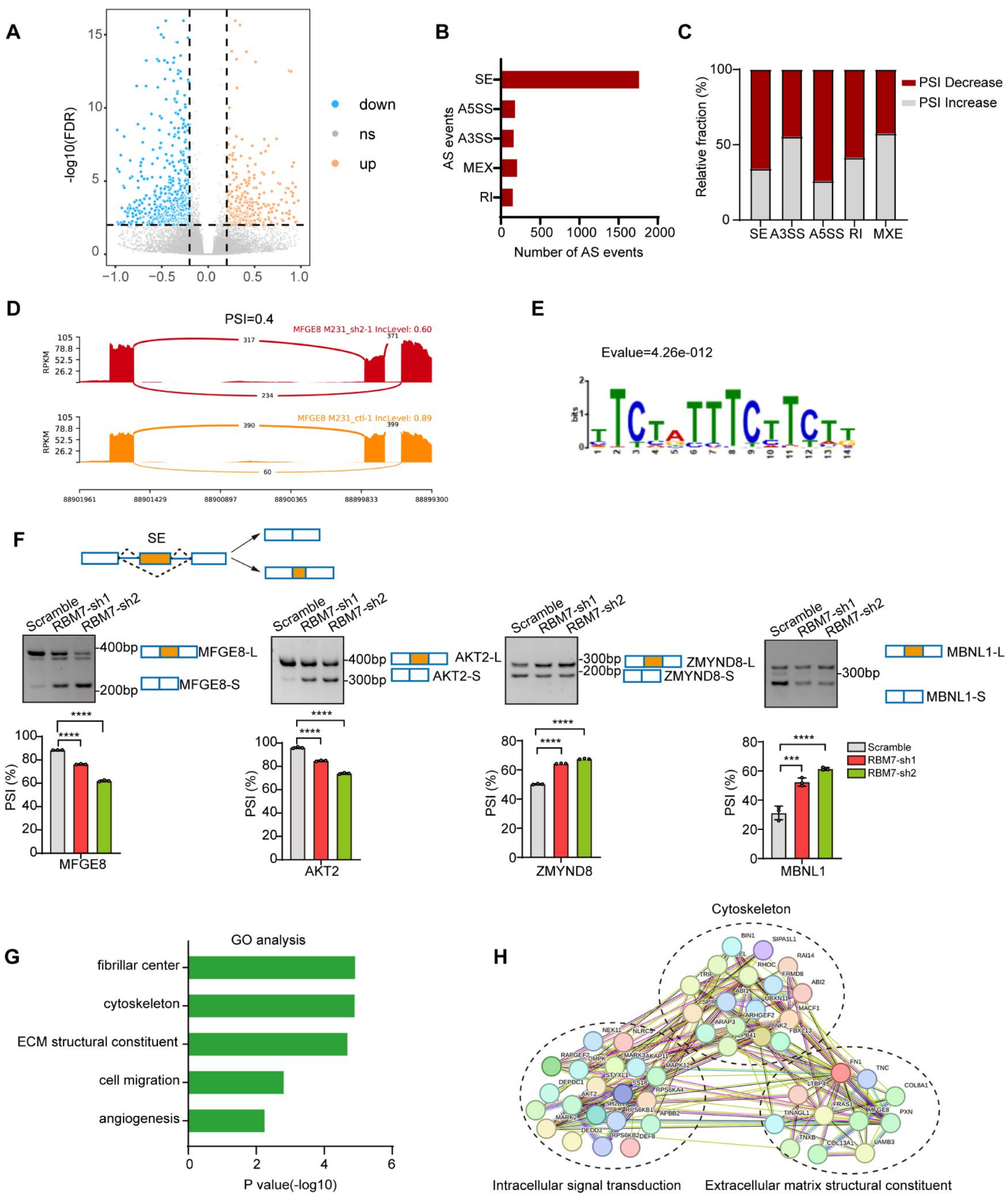
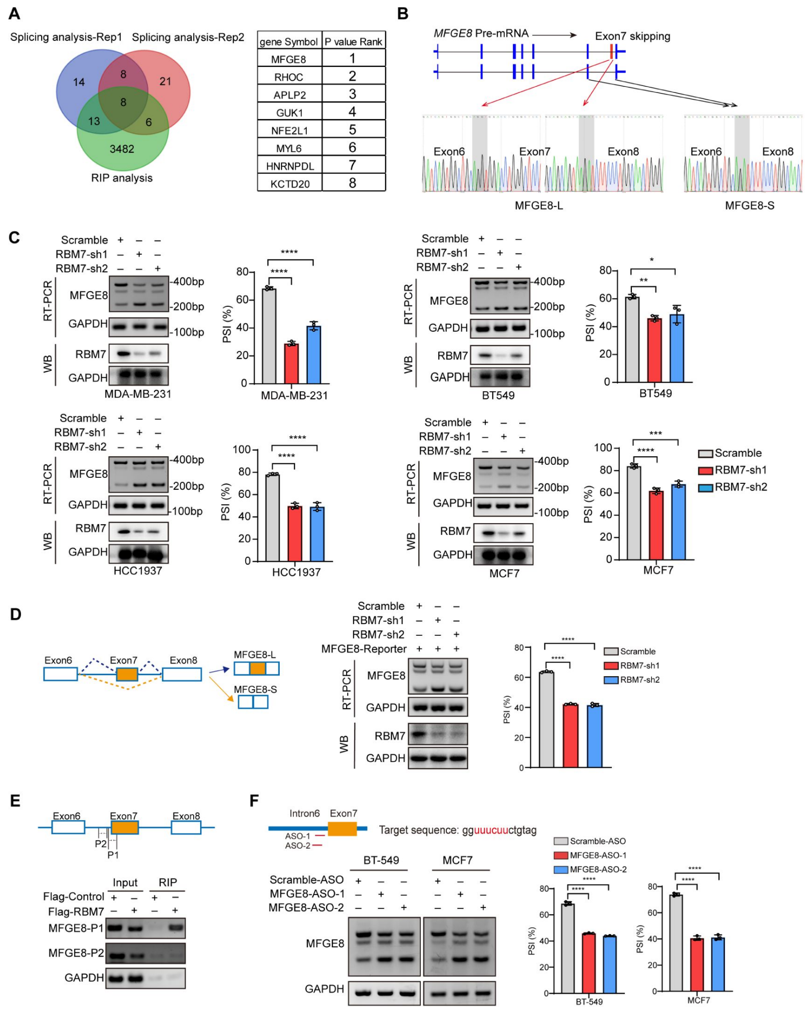


Figure 3.

Global identification of alternative splicing events regulated by RBM7.

(A-E) RNA-seq was performed on RBM7-knockdown MDA-MB-231 cells or control cells. And changes in splicing events were analyzed. (A) Volcano plot illustrating alternative splicing events regulated by RBM7. (B) Quantification of the different AS events affected by RBM7. (C) The relative fraction of each AS event affected positively or negatively by RBM7. (D) Examples of alternative exons affected by RBM7. Alternative splicing of MFGE8 was chosen to represent a decrease of PSI, and numbers of exon junction reads were indicated. (E) The hit top enriched motifs identified in the differentially spliced genes regulated via RBM7. (F) RBM7-regulated exon skipping events were identified by semiquantitative RT-PCR using RBM7-konckdown or control MDA-MB-231 cells. The mean ± SEM from 3 experiments was plotted. P values of percent-spliced-in (PSI) values were determined by one-way ANOVA with Dunnett’s multiple comparisons test. (G) Gene Ontology analysis showed that the significant splicing events affected biological process upon RNM7 knockdown in MDA-MB-231 cells. (H) KEGG pathway enrichment of the RBM7-mediated AS targets.



gene Symbol	P value Rank
MFGE8	1
RHOC	2
APLP2	3
GUK1	4
NFE2L1	5
MYL6	6
HNRNPDL	7
KCTD20	8

Figure 4.

RBM7 knockdown promoted MFGE8 exon 7 skipping via binding to its pre-mRNA.

(A) Venn diagram of 3509 RBM7-binding genes from the RIP-seq (based on data set GSE144075), 43 genes in RBM7-sh1 cells compared with control cells, and 43 genes in RBM7-sh2 cells compared with control cells. The gene lists of the last two groups were from RNA sequencing with top 50 AS events. The gene lists on the right were shown according to the P value. (B) Schematics of human MFGE8 pre-mRNA (NM_005928.4). And MFGE8-long isoform included the exon 7 (hereafter referred to as MFGE8-L), while MFGE8-short isoform skipped the exon (hereafter referred to as MFGE8-S). The arrow indicates the direction of transcription. Primers were designed on the upstream and downstream exons of exon 7, and RT-PCR was performed to identify MFGE8 splicing event using RBM7-KD cells or control MDA-MB-231 cells. The PCR products were subjected to sanger sequencing. The base peak diagram of sanger sequencing showed the splicing junction sites (filled with gray) of upstream and downstream exon 7. (C) Different breast cancer cells lines with stable low-expression of RBM7 or control were constructed. Alternative splicing events of MFGE8 regulated by RBM7 were examined by semi-quantitative RT-PCR. The level of RBM7 protein was detected by Western Blotting assay in various cell lines. The mean ± SD of PSIs from three independent repeated experiments were plotted. P value was calculated by one-way ANOVA with Dunnett's multiple comparisons test. (D) As schematic diagram, an exon of the orange box is a variable exon as shown in the diagram. P1 and P2 were the putative binding regions and primers were designed here. Binding of MFGE8 pre-mRNA with RBM7 was examined by RNA-immunoprecipitation (RIP) in HEK293T cells expressing Flag-RBM7 or Flag-empty plasmid. (E) The diagram shows ASOs targeting the indicated region. MCF7 and MDA-MB-231 cells were transfected with MFGE8-ASO for 48h. And then MFGE8 splicing events were measured by RT-PCR. (F) Schematics of human MFGE8 mini splicing reporter gene and the reporter contained 3 exons and two mini-introns. RBM7 stable knockdown cells or control cells were transfected with mini reporter gene, and RNA was collected 24 hours of transfection. The splicing changes of exogenous MFGE8 were detected by RT-PCR with specific primers on the reporter gene. The level of RBM7 protein was detected in MCF7 cells by Western blotting assay. P value was calculated by one-way ANOVA with Dunnett's multiple comparisons test from three repeated experiments.

The short variant of MFGE8 abolishes the inhibitory function of its classical isoform on breast cancer cell migration and invasion

MFGE8 encodes a secreted protein, Milk fat globule-EGF factor 8 (MFG-E8), which participates in a variety of cellular processes, including mammary gland branching morphogenesis, angiogenesis and mucosal healing, etc (Raymond, Ensslin et al., 2009). We applied MFGE8-L stably overexpressing-or depleted-breast cancer cells to transwell assays and observed the cellular migratory and invasive capabilities was visibly suppressed by MFGE8-L overexpression (Supplementary 4A-4B). On the contrary, the knockdown of MFGE8-L provoked cell migration and invasion (Supplementary 4C). Strikingly, while MDA-MB-231 or HCC1937 expressing MFGE8-L showed a remarkable decrease in migratory and invasive ability compared to control vector, MFG-E8-S not only totally abolished the inhibitory function of MFGE8-L but elicited modest pro-migration/ invasion actions on breast cancer cells (**Figure 5A** and **Supplementary 4D**-**4E**). In comparison with MFG-E8 containing N-terminal EGF-like domain and C-terminal twice-repeated F5/8 type C domains, the protein encoded by MFGE8-S loses the second F5/8 type C domain, implying a shortage of the function corresponding to membrane-binding. Immunofluorescence assays revealed the MFGE8-L forms foci in cytoplasm, particularly enriched near cell-cell contact sites, whereas MFGE8-S is more evenly distributed in the nucleus and cytoplasm (**Figure 5B**). Furthermore, we compared transcriptome profiles between MFGE8-L and MFGE8-S ectopically expressing breast cells and found a set of functional genes were down-regulated by MFGE8-L, whereas this negative regulation was abrogated in response to MFGE8-S (**Figure 5C**). The differentially expressed genes between two splice variants were predominantly enriched in cell adhesion molecules and JAK-STAT cascade as revealed by KEGG pathway and Gene ontology analysis (**Figure 5D**). Subsequent experimental validation demonstrated that the activation of STAT1 was hindered by MFGE8-L, but not affected by MFGE8-S (**Figure 4E**). Considering that STAT1 signaling activation plays an important role in tumor cell metastasis via cell adhesion molecules induction (Marimuthu, Lakshmanan et al., 2022) (Chen, Hao et al., 2019), it is highly plausible that MFGE8-L, but not MFGE8-S, impeded breast cancer cell motility and invasive capability in a STAT1-dependent manner.

RBM7 depletion-stimulated aggressiveness of breast cancer is dependent on MFGE8 splicing switch and NF-kB pathway activation

Next, we sought to assess if MFGE8 splicing switch is responsible for acquisition of aggressive feature in RBM7-silenced breast cancer cells. Flag-tagged MFG-E8-L or an empty vector was co-expressed in MDA-MB-231 cells together with RBM7 shRNA (**Supplementary Figure 5A**). Following seeding into transwell chamber, cells transfected with RBM7 shRNA exerted a higher migratory and invasive capability than scramble control, consistent with above data. Importantly, the restoration of MFGE8-L expression diminished the acquired aggressive properties of MDA-MB-231 cells resulted from RBM7 depletion (**Figure 6A**). Similar results were obtained using RBM4-depleted MCF7 cells with or without MFGE8-L complement (**Figure 6B**). However, a series of shRBM7-upregulated secreted factors that have been reported to drive tumor angiogenesis (Gopinathan, Milagre et al., 2015) (Fousek, Horn et al., 2021) (Feng, Ke et al., 2019), were not impacted by MFGE8-L/S (**Supplementary Figure 5B** and **Figure 6C**). Condition medium derived from breast cancer cells overexpressing MFGE8-L/S showed negligible effect on HUVEC tube formation (**Supplementary Figure 5C**), implying that there exists additional mechanism underlying breast cancer metastasis attributed to RBM7 reduction.

Given that consecutive gene set enrichment analysis (GESA) of highest-ranking gene sets based on transcriptomic profile of mRNA expression showed a signature for NF-κB pathway that was augmented upon depletion of RBM7 (**Figure 6D**), we suspected that the transcriptional activation of downstream genes in NF-κB signaling known to facilitate tumor angiogenesis might

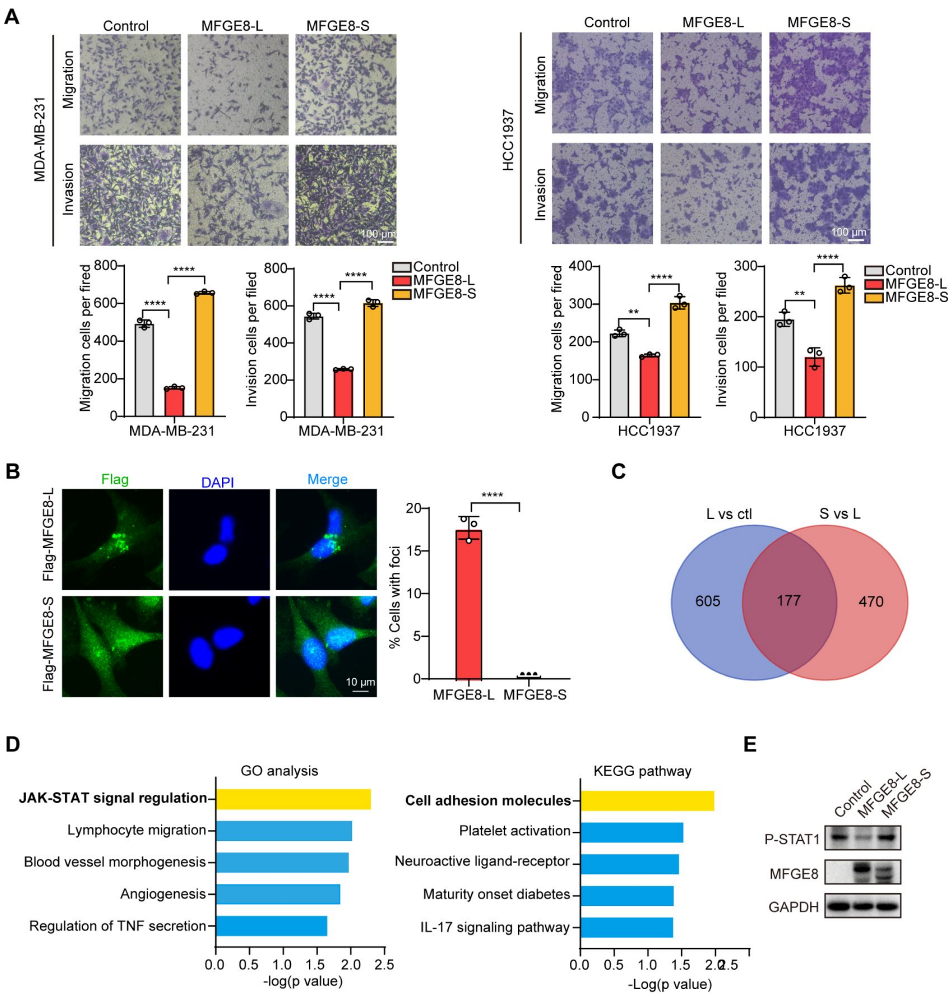


Figure 5.

MFGE8-L inhibited breast cancer cell motility and invasion.

(A) Effects of MFGE8 two isoforms on breast cancer cell migration and invasion were evaluated by Transwell assays. (B) The subcellular localizations of MFGE8 isoforms. MDA-MB-231 cells were transfected with Flag-tagged MFGE8-L or MFGE8-S, and visualized with immunofluorescence assay. Scale bar = 10 μm. The localization of MFGE8-L and MFGE8-S was quantified with foci, and the percent of cells were plotted (about 80 cells were captured and quantified in both samples). (C) Transcriptome sequencing was performed using MDA-MB-231 cells of overexpressing MFGE8-L and MFGE8-S. Venn diagram of 782 MFGE8-L down-regulated genes, 647 genes from MFGE8-S up-regulated genes compared with MFGE8-L. (D) Gene ontology analysis and KEGG pathway analysis of genes up-regulated by MFGE-S compared with MFGE8-L. (E) The expression of MFGE8 and p-STAT1(Tyr701) in HCC1937 cells overexpressing respectively MFGE8-L and MFGE8-S was detected by Western Blotting assay.

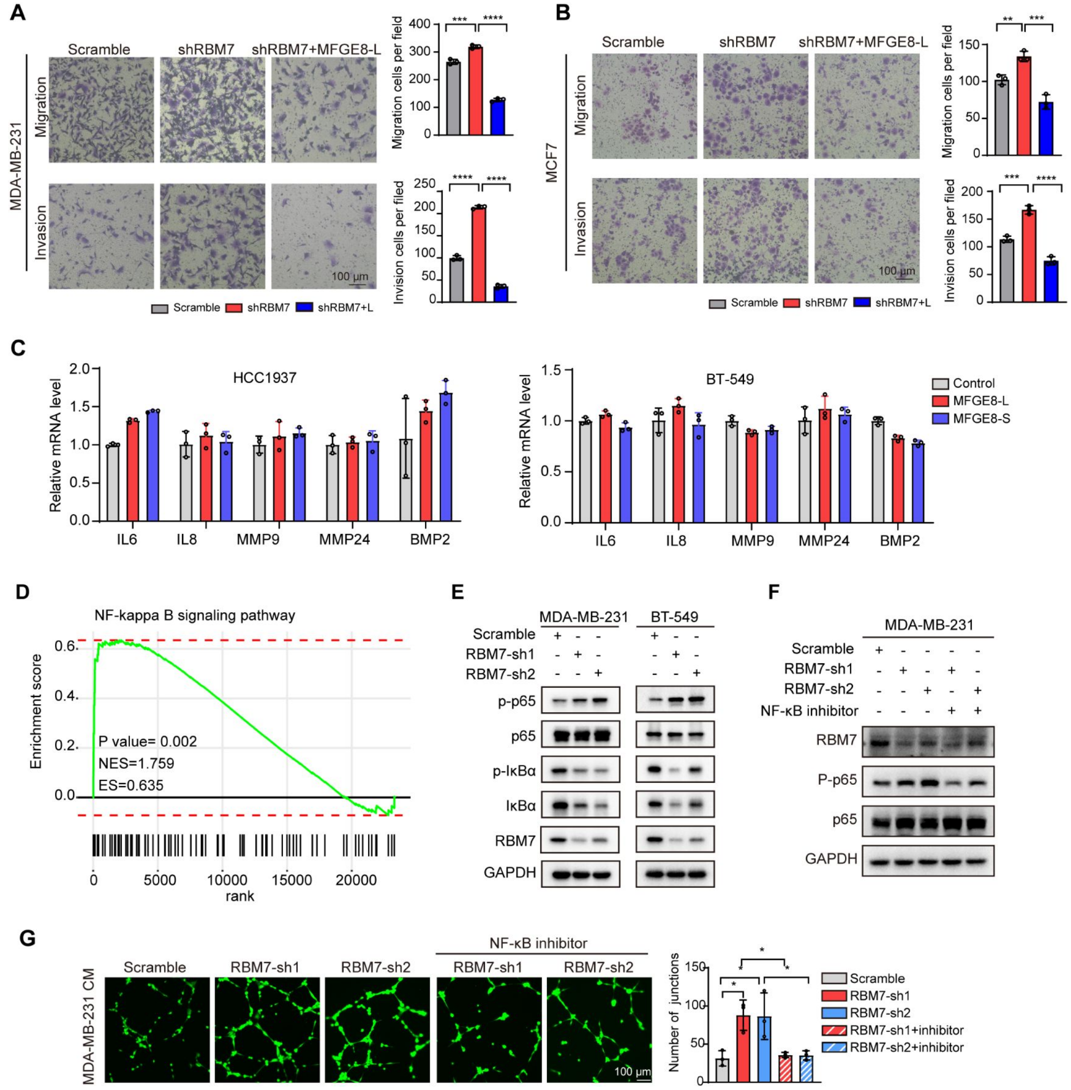


Figure 6.

RBM7 knockdown enhanced aggressiveness of breast cancer relying on MFGE8 splicing switch to the short variant and NF-κB pathway activation.

(A-B) Breast cancer cells MDA-MB-231 (A) and MCF7 (B) were transfected with shRBM7 or control and re-transfection Flag-MFGE8-L vector. Then these cells were subjected to Transwell migration assay or invasion assay. Cells migrating through Transwell membrane with matrigel or not were imaged and representative images are shown (right) along with quantification (left). (C) HCC1937 and BT-549 cells stably expressing MFGE8-L, MFGE8-S and control were generated. The expression of metastasis-related inflammatory factors was measured by real-time PCR assay. (D) NF-κB signaling pathway was enrichment via KEGG analysis in RBM7 knockdown cells. (E) Western blotting showed the expression of NF-κB-associated proteins in RBM7-depleted or control MDA-MB-231 and BT-549 breast cancer cell lines. (F) RBM7-depleted MDA-MB-231 cells were treated with or without NF-κB inhibitor PDTC 10 μm for 48h. Then Western blotting tested the expression of RBM7, NF-κB -p65 and phosphorylated NF-κB p65. (G) Representative images of the tube formation of HUVEC cells, treated with conditional medium with 10 μm NF-κB inhibitor PDTC. Scale bars: 100 μm. Quantification of junctions was by imageJ software.

be involved in shRBM7-exerted pro-metastatic function. As expected, the phosphorylation of p65, a core transcription factor well-known to switch on NF-κB pathway was notably enhanced by RBM7 knockdown, meanwhile, IκBα phosphorylation that can blockade p65 activation was curbed upon RBM7 knockdown (**Figure 6E**). Conversely, RBM7 overexpression impeded the activation of p65 (**Supplementary Figure 5D**). Yet MFGE8-L/S had no effect on phosphorylation of p65 or IκBα (**Supplementary Figure 5E**). We further carried out Matrigel tube formation assays and found NF-κB inhibitor PDTC drastically reversed the promotion of HUVEC angiogenesis by conditional medium harvested from RBM7-depleted cells (**Figure 6F**-**6G**), implying that RBM7-p65 axis may participate in HUVEC angiogenesis through reprogramming the secretome of breast cancer cells. Similar effect was observed with respect to cell motility and invasion (**Supplementary Figure 5F**). Taken together, these data above indicated that both MFGE8 splice switch and NF-κB pathway activation are required for mediating the pro-metastasis activity of RBM7 reduction in breast cancer.

The PSI of MFGE-8 exon 7 is decreased in breast cancer patients and positively correlated with RBM7 expression

To study the clinical relevance of MFGE-8 isoform switch, we analyzed the PSI changes of cassette exon7 in TCGA breast cancer cohort and found exon7 skipping occurs more frequently in tumor than in normal tissues (**Figure 7A**). Consistently, in comparison with normal counterparts, the splicing of MFGE8 is shifted towards a reduced proportion of MFGE8-L across multiple cancer types, including LUAD, HNSC, COAD, etc (Supplementary 6A). RT-PCR assay was performed in surgical tumor specimens in breast cancer patients and showed MFGE8 exon7 is prone to skipping in tumor (**Figure 7B**). On the other hand, the splicing switch in lymph node metastases favored MFGE exon7 exclusion when compared to paired primary tumor tissues from 3 breast cancer patients (**Figure 7C**). This is coherent with decreased RBM7 expression levels found in breast cancer with lymph node metastasis. In addition, patients with high expression of MFGE8 exon7 had a visibly better overall survival than patients with low expression (**Figure 7D**). We performed Pearson correlation test and revealed a noticeably positive correlation between the expression of RBM7 and exon7 inclusion of MFGE8 in breast cancer (**Figure 7E**), further supporting MFGE8 isoform switch regulated by RBM7.

Discussion

Previous reports have recognized the RNA-binding protein RBM7 as a component of the NEXT complex that could facilitate RNA exosome targeting to a subset of non-coding RNAs and aberrant transcripts (Meola et al., 2016) (Pascal Preker, 2008), including short-lived exosome substrates and pre-mRNAs, for surveillance and turnover. In addition to this specific characteristic, RBM7 was shown to interact with subunits of 7SK snRNP complex and thus activate P-TEF downstream genes in response to DNA damage (Bugai, Quaresma et al., 2019). Herein, we discover a novel role of RBM7 in alternative splicing control through directly binding to its cognate cis-element in pre-mRNA. Using whole-transcriptome sequencing, we found the depletion of RBM7 contributed to exon exclusion in approximately 70% SE/A5SS events, implying RBM7 mainly functions as a splicing activator in breast cancer. However, the increase of PSI also occurred due to RBM7 knockdown in a smaller faction of splicing events, such as ZMYND8. We speculated this converse impact may be associated with the intrinsically sophisticated splicing processes driven by distinct core splicing factors, cognate cis-acting element and co-factors in a context-dependent manner (e.g., pre-mRNA structure, base modification, antagonistic or cooperative effect of multiple splicing factors). Similar to many splicing factors, like SR proteins and hnRNPs, RBM7 has an RRM-domain required for target RNA recognition in the N-terminus. Through ectopic expression of a series of RBM7 truncated mutant, we unveiled the disordered region (162-266) rich in polar residues serves as the functional modular to execute alternative splicing program.

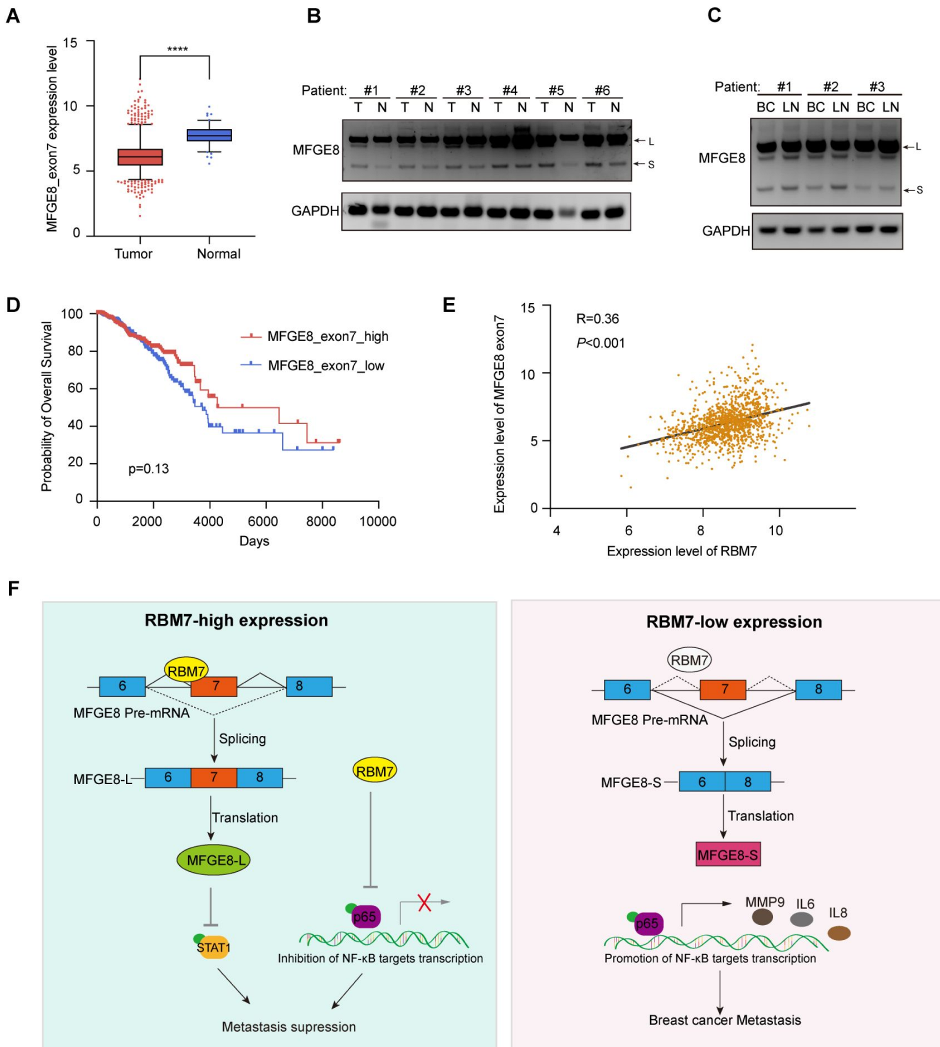


Figure 7.

Splicing shift of MFGE8 toward exon7 exclusion was negatively correlated with RBM7 expression in patients with breast cancer.

(A) MFGE8_exon7 expression level was analyzed in breast carcinoma from the TCGA dataset. (B-C) MFGE8 alternative splicing events were quantified in 6 pairs of primary BRCA tissues and adjacent normal tissues using RT-PCR (B). Semi-quantitative RT-PCR was performed using 3 pairs of primary breast cancer tissues and lymph node metastasis of breast cancer tissues (C). (D) Correlation of RBM7 with the MFGE8 exon7 expression levels was analyzed with Pearson's correlation coefficient. (E) The correlation between MFGE8-S and overall survival (OS) of breast cancer patients based on the TCGA dataset. (F) The mechanistic model of this study. Schematic illustration of the roles of RBM7-MFGE8 splicing switch axis in metastasis of breast cancer.

Currently, the pathobiological function of RBM7 in breast cancer is underexplored except one study that claimed an oncogenic role of RBM7 since its knockdown inhibited breast cancer cell proliferation (Xi et al., 2020). Nevertheless, in our study, RBM7 knockdown in breast cancer displayed visibly pro-metastasis function with activated NF-κB signaling. More importantly, according to TCGA database (total cases > 1000 used for analysis), RBM7 shows decreased expression in breast cancer, particularly in the metastasis loci and has positive correlation with patients' survival outcomes. It is therefore highly doubtful that RBM7 could serve as an oncogene in breast cancer as previously reported. Moreover, we can't exclude additional biological function of RBM7 in breast cancer. RBM7-targeted silencing for breast cancer would likely be detrimental for neighbouring heathy tissue and immune environment as a result of NF-κB-dependent pro-inflammatory mediators and cytokines secretion. Other splicing events switch triggered by RBM7 might also be linked to tumor development. For example, we found RBM7 regulated alternative splicing of AKT2, which is involved in glucose metabolism (Hay, 2011). Further investigation is warranted to elaborate more splicing mechanisms of RBM7 in breast cancer.

In conclusion, we discovered that down-regulated RBM7 and its cognate elevated exon7-exclusion MFGE8 were tightly related to enhanced breast cancer metastatic potential and poor prognosis of patients. MFGE8-L, the canonical variant, possessed the capability to trigger STAT1 phosphorylation and inhibit the motility and invasion of breast cancer cells, whereas splice switch to MFGE8-S was found loss-of-function for this tumor-suppressive role. At another layer, RBM7 reduction led to NF-κB pathway activation, resulting in enhancement of breast tumor aggressiveness. Overall, our findngs implicated that activated NF-κB pathway worked in concert with MFGE8 splicing transition to spur metastasis in RBM7-depleted breast cancer. Additionally, although the detailed clinical significance underlying their cooperation need further in-depth investigation, we propose that RBM7 and its regulated splicing variants might serve as potential therapeutic biomarker and prognostic indicator in breast cancer patients.

Materials and Methods

Cell culture

Human breast cancer cell lines (MDA-MB-231, MCF7, HCC1937 and BT-549) and human embryonic kidney cell HEK293T in this article were obtained from the American Type Culture Collection (ATCC). MDA-MB-231 cells were cultured without carbon dioxide at 37℃ in Leibovitz's L-15 medium (Gibco) contained 10% FBS (SINSAGE TECH). MCF7 and HEK293T cells were cultured with DMEM mediun (Gibco) supplemented with 10% FBS. HCC1937 cells were cultured with RPMI-1640 media (Gibco) supplemented with 10% FBS, while BT-549 was maintained with RPMI-1640 media (Gibco) supplemented with 10% FBS and 0.023 U/ml insulin. HUVECs were cultured with Endothelial Cell Medium (ScienCell) supplemented with ECGS (ScienCell), P/S solution (ScienCell) and 10% FBS (ScienCell). Except for MDA-MB-231, all the cells above were cultured in a 37 °C humidified incubator.

Constructions of stable cell lines

To generate RBM7 overexpression vector, the coding regions of this gene (NM_016090.4) were cloned into pCDH-CMV-MCS-EF1-Puro with N-terminal Flag tag. For RBM7 knockdown, target-specific shRNA (RBM7-sh1: 5'-CGAAAGGACTATGGATAACAT-3';RBM7-sh2:5'-CTACCTAGCAGATAGACATTA-3') was cloned into pLKO.1 vector. And HEK293T cells were transfected these over-expression (or low-expression) lentivirus plasmis together with lentivirus packaging plasmids pPAX2 and pMD2 according to the manufacturer's instructions. Viruses were collected 48 hours after transfection and then to infect breast cancer cell line with 8 ug/ml polybrene (beyotime biotechnology) for 24 hours. Positive cells were selected with 3 ug/ml puromycin.

Western Blot assay

Cells were lysed with RIPA buffer (10 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.1% SDS, 1% Triton X-100, and 1% sodium deoxycholate) supplemented with protein tyrosine phosphatase inhibitor Na_3VO_4 (1mM), serine protease inhibitor PMSF (1mM) and protease inhibitor Cocktail. Immunoblotting assay was performed using standard methods. The following antibodies were used: anti-RBM7 (Sigma-Aldrich, HPA013993), anti-Flag (Sigma-Aldrich, F1804), anti-GAPDH (Proteintech, 10494-1-AP), anti-Phospho-Akt (Ser473) (Cell Signaling Technology, #4060), anti-Akt (pan) (Cell Signaling Technology, #4691), anti-Phospho-NF-κB p65 (Cell Signaling Technology, #3033), anti-NF-κB p65(Cell Signaling Technology, #8242), anti-IκBα (Cell Signaling Technology, #4814), anti-Phospho-IκBα (Ser32) (Cell Signaling Technology, #2859), anti-Phospho-stat1 (Tyr701) (Cell Signaling Technology, #8062),and anti-MFGE8(R&D systems, MAB2767). Used secondary antibodies were conjugated to horseradish peroxidas. The protein bands were detected with an ECL Enhanced chemiluminescence regent Kit (NCM, Biotech), visualized using the MiniChemi™ Cheminescent image system (Beijing Sage Creation, China).

RNA isolation, RNA sequencing, and PCR

Total RNA from cells and tumor tissues was extracted using Trizol (Invitrogen) according to the manufacturer's instructions. Total mRNA was cleaned up using RNeasy Lipid Tissue Mini Kit (Qiagen, USA). And RNA-seq was carried out on platform of Novogene Illumina sequencing. cDNAs were obtained using *Evo M-MLV* Plus 1st Strand cDNA Synthesis Kit (Accurate Biotechnology, AG11615) with random primers. qRT-PCR was carried out using MonAmp^{TM} Universal ChemoHS Specificity Plus qPCR Mix. GAPDH mRNA was examined as control for normalization. Gene expression relative to GAPDH was calculated using the formula $2^{-\Delta\Delta CT}$. For identified alternative splicing events, primers of the upstream and downstream of variable exons or introns were designed and PCR assay was performed. PCR products were subject to DNA agarose gel electrophoresis. AS events change of percent-spliced-in (PSI) values were the ratio of retained splices to total transcripts. The amount of each splicing isoform was measured with ImageJ. All the primers are showed in Table1.

Migration and invasion assays

For transwell migration assay, 10×10^4 MDA-MB-231 or other breast cancer cells (MCF7, BT-549 and HCC1937) were plated in the upper chamber of 6.5 mm insert (Costar, USA) with 8.0 μm polycarbonate membrane with 100 μL serum-free medium. For invasion assay, 10×10^4 breast cancer cells were seeded in the top chamber of each Matrigel-coated insert (BD Biosciences, USA). After 24 hours or 48 hours of incubation at 37 °C, migrated or invaded cells were fixed with 4% PFA and stained in 0.1% crystal violet. The cell number was counted by ImageJ software in 3 views from 3 independent experiments.

Wound healing

Knockdown RBM7 in MDA-MB-231 and 4T1 cells or control cells were implanted into 6-well culture dishes. When the cells grew about 90% confluence, lines of the same width were drawn on the bottom of the culture dish using autoclaved sterile tips. Images were captured at 0 and 36 hours after the wounding. Data shown were representative of three independent repeats.

Tube formation of HUVECs assay

Capillary tube formation assay was used to observe angiogenesis of HUVECs. HUVECs were seeded into 48-well plate coated with growth factor reduced Matrigel (BD Biosciences). HUVECs cells were treated with conditioned medium for 4-5 hours and then formation of capillary-like tubes was observed and stained with calcein for 30 minutes. Images of blood vessels were taken using fluorescence microscopy. And the tube length was analyzed by ImageJ software.

Immunofluorescence assay

After seeded on coverslips for 36 hours, the MDA-MB-231 cells were immersion fixed with 4% PFA for 20 minutes at room temperature, washed 3 times in PBS. Then the cells were treated with 0.2 % Triton-X100 in PBS for 10 minutes and blocked with 3% BSA for 10 minutes. Next the cells were immersed in primary antibody at 4℃ overnight, washed in PBS 3 times for 5 minutes each, and incubated with secondary antibody labelled with Alexa fluorophore 488/592 for 1 hour at RT. After washing 3 times in PBS, the cells were dyed with DAPI. In addition, antibodies targeting RBM7 (Sigma, HPA013993), and Flag (Sigma, 3165) were used in the assay. The pictures are captured with fluorescent microscope (Leica, USA).

RNA immunoprecipitation assay

RNA immunoprecipitation was performed as previously described. In brief, HEK293T cells were transfected Flag-RBM7 or control plasmid. After 48 hours of transfection, Flag-RBM7 or control cells were collected and cross-linked by 1% formaldehyde, homogenized in lysis buffer (50 mM Tris-HCl Ph 7.5, 0.4 M NaCl, 1 mM EDTA, 1mM DTT, 0.5% TritonX-100, 10% Glycerol supplemented with RNase inhibitor and broad-spectrum protease inhibitors Cocktail), and broke with an ultrasonic crusher. And solubilized cell lysate was obtained and precleared by ProteinG-agarose beads together with nonspecific tRNA to remove nonspecific binding. Then binding RNAs with RBM7 protein were precipitated with Anti-Flag M2 Affinity beads. Co-precipitated RNAs were subject to RT-PCR. Input controls (total RNAs) and Flag-controls were assayed simultaneously to prove that the detected signals were the result of RNAs specifically binding to RBM7. The gene-specific primers were designed MFGE8-RIP1-F:5’-tatttatgcccccctcaccgc-3’; MFGE8-RIP1-R:5’-atggtggctgcctgtaat-3’;MFGE8-RIP2-F:5’-ctgaggctcacccaagtag-3’;MFGE8-RIP2-R:5’-GTAGGATGCCACAAACTG-3’). GAPDH mRNA was as negative control (GAPDH-F 5’-GGGGAAGGTGAAGGTCG-G-3’; GAPDH-R 5’-TTGAGGTCAATGAAGGGG-3’).

Antisense oligonucleotide (ASO)-mediated splicing blockade

MFGE8-ASOs were synthesized with a full-length phosphorothioate backbone and 2’-O-methyl modified ribose molecules (GenScriptBiotech Corp). ASO was transfected at final concentration of 100 nM using Lipofectamine3000 (Thermo Fisher Scientific) according to the manufacturer’s instructions. The MCF7 cells were harvested for RNA and splicing assays 48 h after transfection.

Animal experiments

All procedures and experimental protocols involving mice were approved by the Institutional Animal Care and Use Committee (IACUC) of Dalian Medical University. Male NYG mice (6 weeks old, 4-5 mice per group as indicated in figures and/or figure legends) were injected into tail vein with murine 4T1 breast cancer cells (6×10^5 cells) to develop mice model of pulmonary metastasis. After about 2 weeks, all mice were euthanized, and lungs were harvested, fixed and paraffin embedded. After lungs were H&E stained, the number of metastatic lesions were enumerated by pathologist using brightfield microscopy.

Clinical tissues samples collection

All human breast cancer tissues or normal tissue were obtained with written informed consent from patients or their guardians prior to participation in the study. The Institutional Review Board of the Dalian Medical University approved use of the tumor specimens in this study.

Statistical analyses

Statistical analyses and statistical graphs were conducted by GraphPad Prism 8 software. To compare the differences between the two variables, Student’s T-test, one-way ANOVA with Dunnett multiple comparisons, and one-way ANOVA with Tukey’s multiple comparisons according to the specific situation were applied. *p < 0.05, **p < 0.01, ***p < 0.001, and ****p < 0.0001 indicated the significant difference.

Data availability

The authors declare that all the data supporting the findings of this study are available within the article and its supplementary information files. RNA-seq data in this study have been deposited in Gene Expression Omnibus of NCBI with the accession code GSE248938.

Author contributions

YW and YQ conceived the project and designed the experiments. FH, JY, ZD, KW, and DC performed most of the experiments, whereas CC, JZhang, JZhao and WZ performed data analysis. YW and YQ wrote the manuscript.

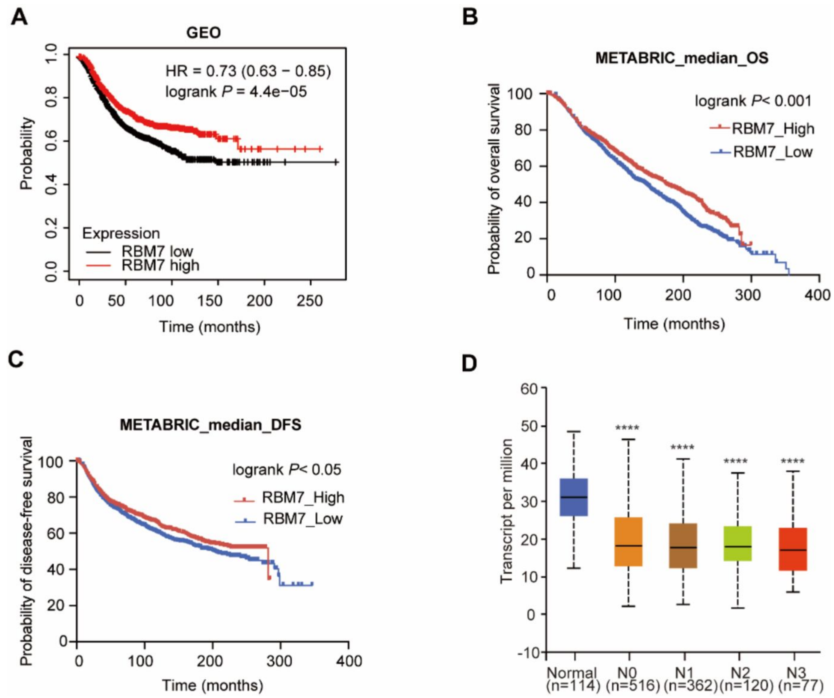
Acknowledgements

This work was supported by the National Key R&D Program of China (2022YFA1104002 to YW); the National Natural Science Foundation of China (82225034, 81830088 to YW; 82103148 to YQ; 81872247 to WZ); the Science and Technology Innovation Foundation of Dalian (2022JJ11CG009 to YW), Liaoning Revitalization Talents Program XLYC1802067 to YW; the Department of Science and Technology of Liaoning Province (2021JH6/10500160 to YW); the Basic Scientific Research Project of Education Department of Liaoning Province (LJKQZ2021104 to YQ); the Science and Technology Innovation Talent Support Program of Dalian (2022RQ056 YQ); Dalian High Level Talents Renovation Supporting Program (2019RQ097 to WZ).

Study Approval

The Institutional Animal Care and Use Committee of the Dalian Medical University approved use of animal models in this study. All human tumor tissues were obtained with written informed consent from patients or their guardians prior to participation in the study. The Institutional Review Board of the Dalian Medical University approved use of the tumor specimens in this study

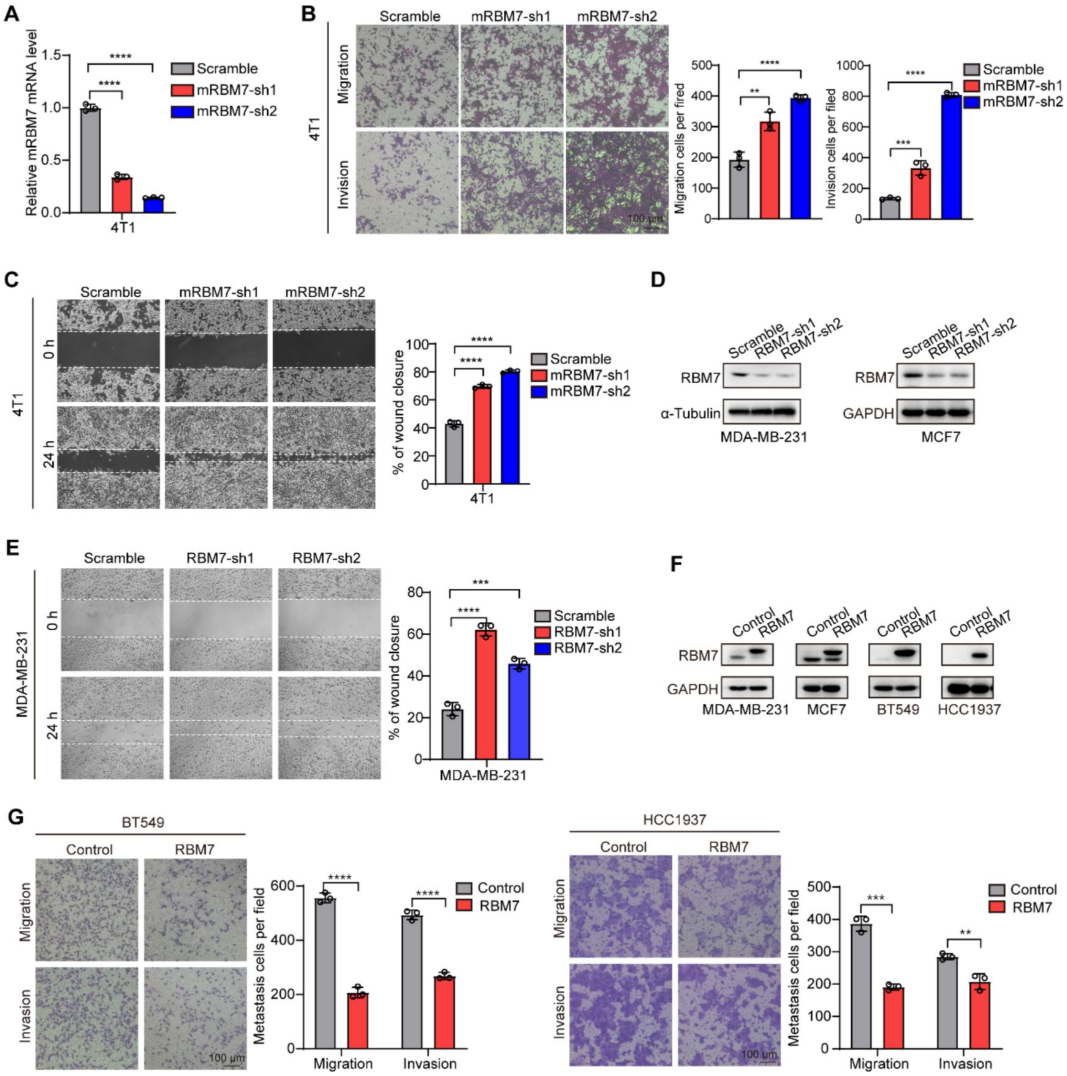
Primers	Gene	primer name	sequence
Primers for qRT-PCR	RBM7	RBM7-qF	5'-AAGCGGATCGCACTCTCTTT-3'
		RBM7-qR	5'-ACGCAAACTGCTTTGGTTTAC-3'
	GAPDH	GAPDH-qF	5'-ATGGGGAAGGTGAAGGTCG-3'
		GAPDH-qR	5'-GGGGTCATTGATGGCAACAATA-3'
	MFGE8	MFGE8-qF	5′-TTCCCAAGAAGTGCGAGGAGATGT-3'
		MFGE8-qR	5′-ATGCTGCAAACCCAAGAAGGTCAC-3'
	IL6	IL6-qF	5′-GAAAGCAGCAAAGAGGCAC-3'
		IL6-qR	5′-GCTCTGGCTTGTTCCTCACTAC-3'
	BMP2	BMP2-qF	5'-TCGCAGGCACTCAGGTCAG-3'
		BMP2-qR	5'-TTCCCACTCGTTTCTGGTAGTT-3'
	MMP9	MMP9-qF	5'-AACTTTGACAGCGACAAGAAGT-3'
		MMP9-qR	5'-ATTCACGTCGTCCTTATGCAAG-3'
	IL8	IL8-qF	5'-GTGCAGTTTTGCCAAGGAGT-3'
		IL8-qR	5'-CTCTGCACCCAGTTTTCCTT-3'
	MMP24	MMP24-qF	5'-ATCTGCTTCCCTATGACTCACG-3'
		MMP24-qR	5'-CGCTTGTTTCTCCGCCTAC-3'
	mGAPDH	mGAPDH-qF	5'-CCTTCCGTGTTCCTACCCC-3'
		mGAPDH-qR	5'-GCCCTCAGATGCCTGCTTC-3'
	mRBM7	mRBM7-qF	5'-GTGTCTGTTCCCTATGCC-3'
		mRBM7-qR	5'-CACGTTACCCACTGTCCT-3'
Primers for alternative splicing	MFGE8	MFGE8-SE-F	5'-TGGGCCTGAA GAATAACA-3'
		MFGE8-SE-R	5'-CAAGTTCTTCTTGTGGGA-3'
	AKT2	AKT2-SE-F	5'-GCGTGTCTTCACAGAGGAGC-3'
		AKT2-SE-F	5'-AGCCCAGCAAGCAGGGACT-3'
	ZMYND8	ZMYND8-SE-F	5'-ACCCTTGACCTTTCTGGCT-3'
		ZMYND8-SE-R	5'-CCAGCCGAGACTCTTTCG-3'
	MBNL1	MBNL1-SE-F	5'-TACCAACGTG GCAATTGC-3'
		MBNL1-SE-R	5'-TAATGGGGGAAGTACAGCTT-3'
	MAP7D1	MAP7D1-RI-F	5'-ATCGCAGCCTGCAGCTGA-3'
		MAP7D1-RI-R	5'-TGCACGCTTCGGGTGACGCT-3'
	HNRNPC	HNRNPC-A5E-F	5'-GGCTTTGCCTTCGTTCAG-3'
		HNRNPC-A5E-R	5'-CATCCTATCATAATAGTCCCGTTG-3'



Supplement Figure1.

The low expression of RBM7 in breast cancer was positively correlated with the poor survival of patients.

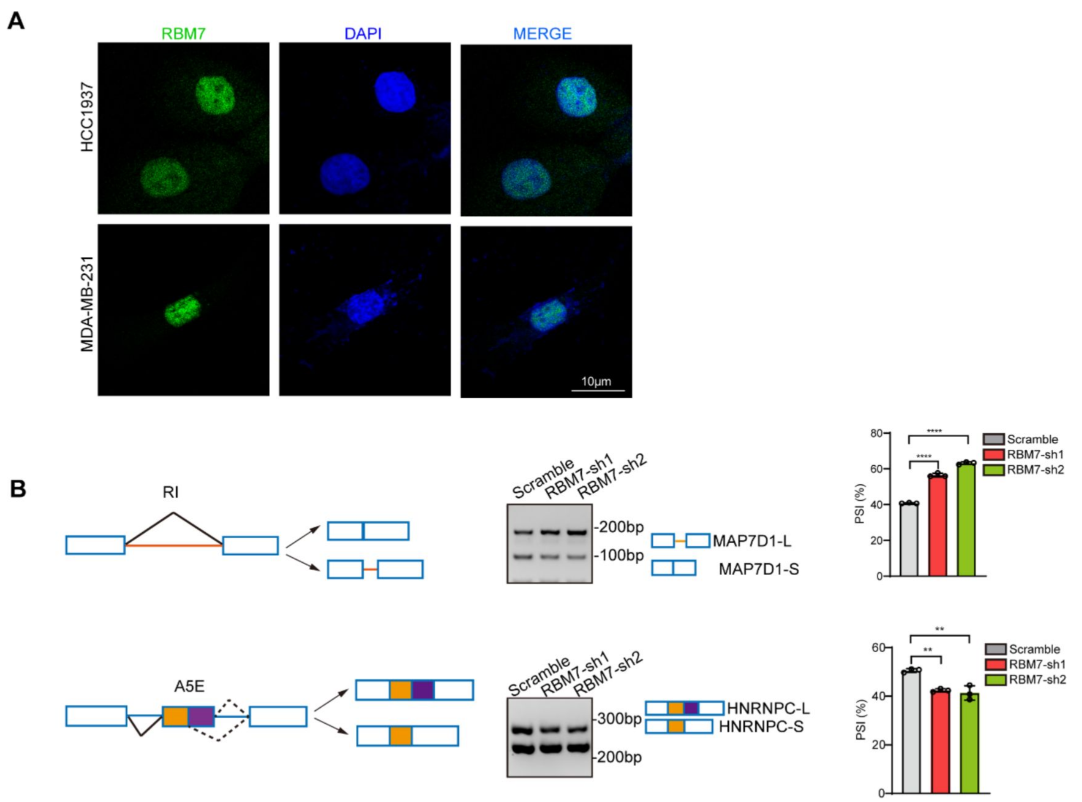
(A)The correlation between RBM7 expression and overall survival of breast cancer patients based on data from the Kaplan–Meier Plotter cohort. (B-C) METABRIC analysis of the correlation between RBM7 expression and overall survival (OS) and disease-free survival (DFS) of breast cancer patients based on the METABRIC dataset. The p value was obtained by log-rank test (A, B, C). (D) RBM7 mRNA level was analyzed in breast invasive carcinoma based on nodal metastasis status from the UALCAN dataset. The p-value was calculated by unpaired Student's t-test by comparing lymph node metastasis of breast cancer tissues with normal tissue.



Supplement Figure2.

RBM7 inhibited breast cancer cell migration and invasion.

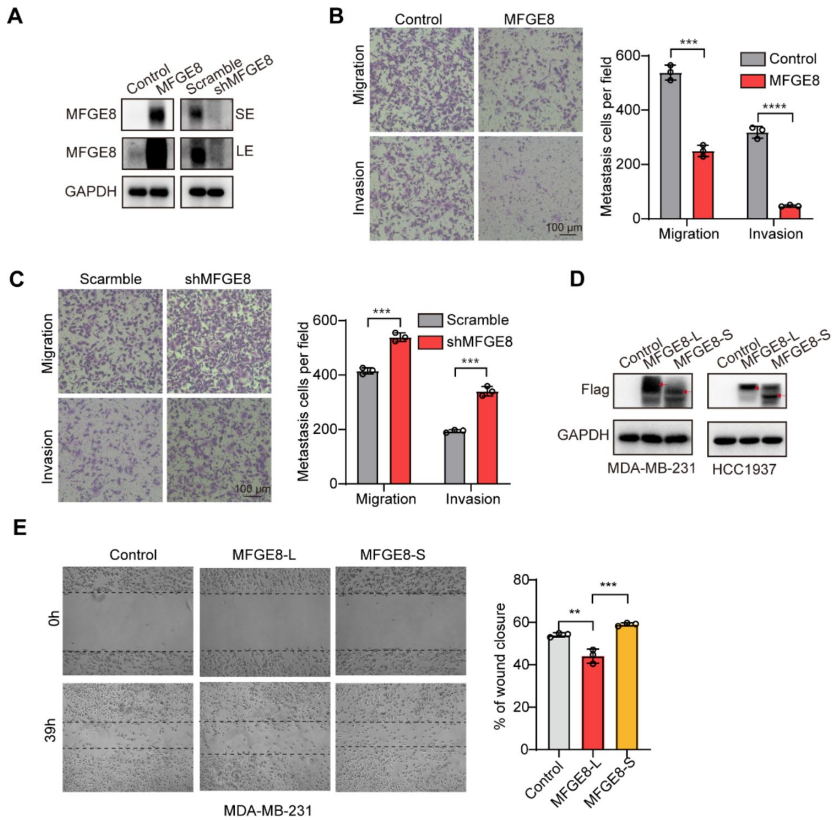
(A) The mouse cell line with RBM7 knockdown was constructed. And the expression of RBM7 was detected by real-time quantitative PCR. (B) The migration and invasion ability of murine RBM7 was detected by Transwell assay using RBM7-KD 4T1 cells. (C) The metastatic capability of RBM7 on 4T1 cells was evaluated by wound healing assay. (D) Western blotting showed the expression of RBM7 in MDA-MB-231 cells and MCF7 cells of RBM7-depleted. (E) The metastatic capability of RBM7 on MDA-MB-231 cells was evaluated by wound healing assay. (F) Western blotting showed the expression of RBM7 in various breast cancer cell lines of RBM7 over-expression. (G) The metastatic ability of RBM7 for BT-549 cells and HCC1937 cells of promoted expression of RBM7 was evaluated by Transwell assay. Data are mean ± SD from three random fields. P values were determined by unpaired Student's T test (G). P value was calculated by one-way ANOVA with Dunnett's multiple comparisons test (A, B, C, E).



Supplement Figure3.

RBM7 could regulate alternative splicing in breast cancer cells.

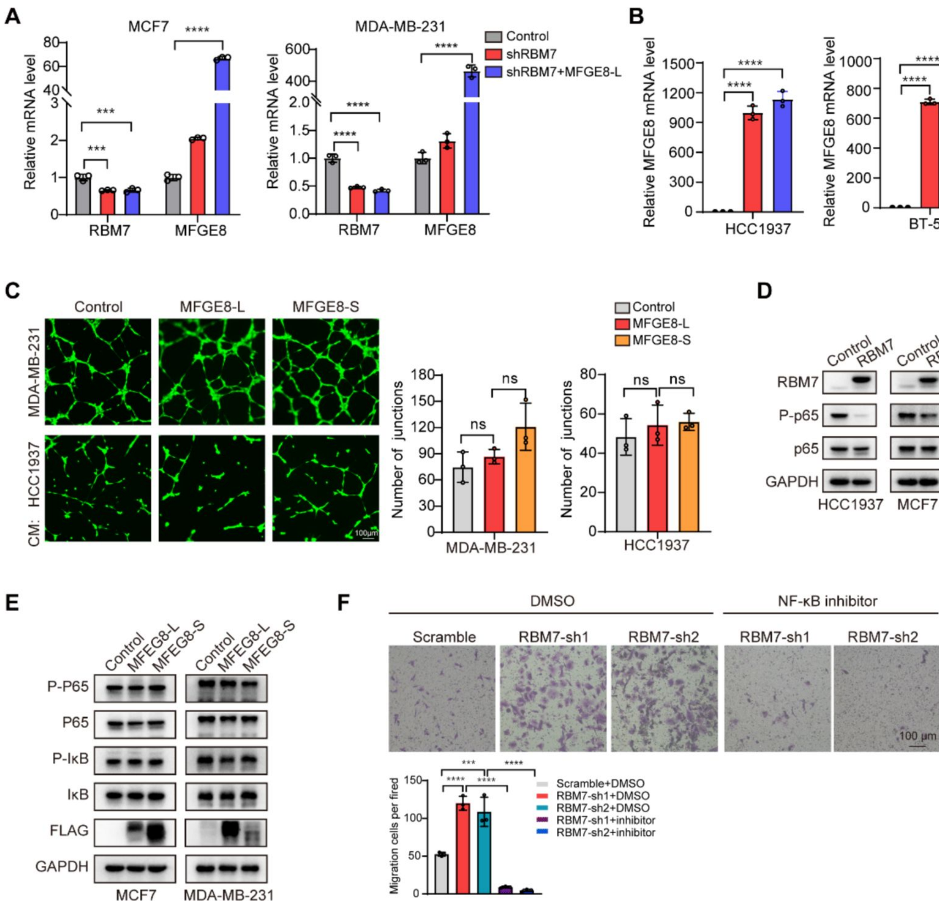
(A) Representative image of MDA-MB-231 and HCC1937 cells stained for RBM7. Scale bar=10 μm. (B) Semi-quantitative RT-PCR was performed to validate the AS events in MAP7D1, and HNRNPC in MDA-MB-231 cells with RBM7 knockdown compared with controls. P value was calculated by one-way ANOVA with Dunnett's multiple comparisons test.



Supplement Figure4.

MFGE8-L suppressed breast cancer cells migration and invasion.

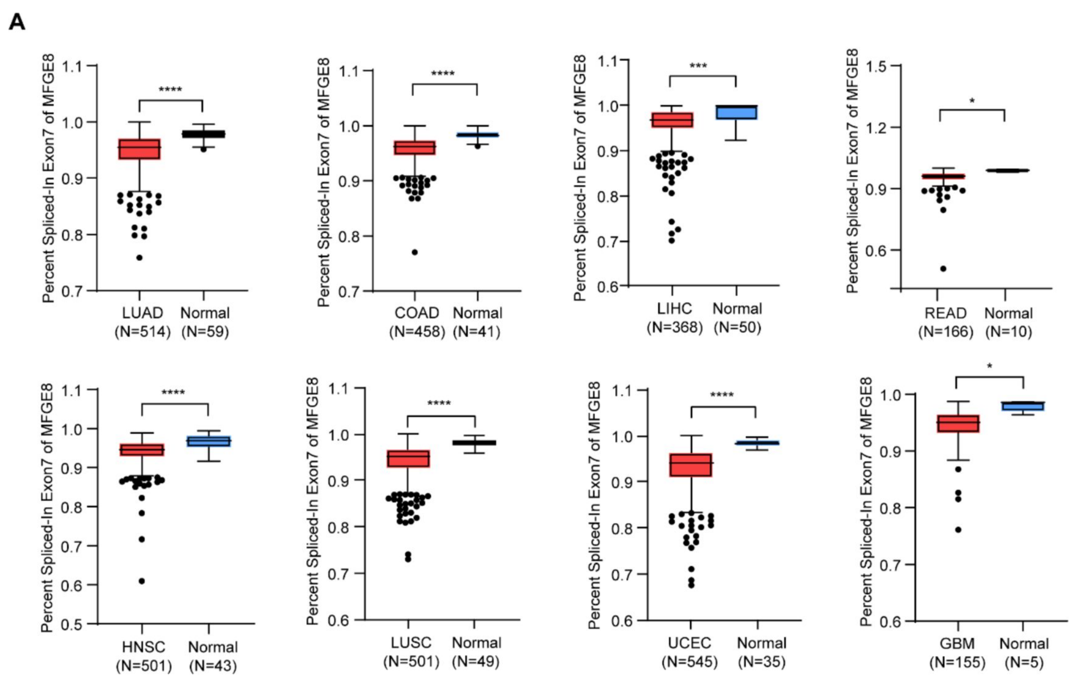
(A) The protein expression of endogenous MFGE8 was determined by western blot in MDA-MB-231 cells knocking down or over-expression MFGE8. SE denotes long exposure and LE denotes long exposure. (B-C) Transwell assay with Matrigel or not was performed using MFGE8 promoted MDA-MB-231 cells or MFGE8 decreased expression cells. P value was calculated by unpaired Student's T test. (D) Western blot showed the extrinsic expression of MFGE8-L isoform and MFGE8-S isoform in MDA-MB-231 and HCC1937 transfected respectively Flag-MFGE8-L, Flag-MFGE8-S and empty vectors. The red arrow indicates the location of the protein. (E) MDA-MB-231 cell line expressing MFGE8-L, MFGE8-S or a vector control were analyzed by wound healing assay. Percent of wound closure was measured in triplicate experiments, with mean ± SD plotted (p values were calculated by one-way ANOVA with Tukey's multiple comparisons test).



Supplement Figure5.

MFGE8-L/S isoforms had no effect on NF-κB pathway in breast cancer cells.

(A) MCF7 and MDA-MB-231 cells stably expressing shRBM7, shRBM7/MFGE8 and control were generated. The expression of RBM7 and MFGE8 was measured by real-time PCR assay. The mean ± SD of relative mRNA levels from triplicate experiments are plotted. (B) HCC1937 and BT-549 cells stably expressing MFGE8-L, MFGE8-S and control were generated. The expression of MFGE8 was measured by real-time PCR assay. (C) Representative images of the tube formation of HUVEC cells, treated with conditional medium, which cultured MFGE8-L, MFGE8-S stably transfected cells and control cells for 48h and collected. Scale bars: 100 μm. Quantification of junctions was by imageJ software. (D) Western blotting showed the expression of indicated proteins in RBM7-promoted or control breast cancer cells. (E) Western blotting showed the indicated antibodies in up-expression of MFGE8 different isoforms in breast cancer cells. (F) RBM7-depleted MDA-MB-231 cells were treated with or without NF-κB inhibitor PDTC 10 μm for 48h. Then Transwell assays were performed to assess cell migration ability. p values were calculated by one-way ANOVA with Tukey's multiple comparisons test (C, F).



Supplement Figure6.

MFGE8 exon7 skipping is enhanced in a variety of cancers.

(A) MFGE8_exon7 skipping event was analyzed in various cancers according to TCGA SpliceSeq dataset. The p-value was calculated by unpaired Student's t-test.

References

Anczuków O *et al.* (2015) **SRSF1-Regulated Alternative Splicing in Breast Cancer** *Molecular Cell* **60**:105–117

Baralle D (2005) **Splicing in action: assessing disease causing sequence changes** *Journal of Medical Genetics* **42**:737–748

Bates SJHaDO (2008) **VEGF-A splicing: the key to anti-angiogenic therapeutics?** *NATURE REVIEWS* **7**

Bugai A *et al.* (2019) **P-TEFb Activation by RBM7 Shapes a Pro-survival Transcriptional Response to Genotoxic Stress** *Molecular Cell* **74**:254–267

Chen C, Zhao S, Karnad A, Freeman JW (2018) **The biology and role of CD44 in cancer progression: therapeutic implications** *Journal of Hematology & Oncology* **11**

Chen Y, Hao X, Sun R, Wei H, Tian Z (2019) **Natural Killer Cell–Derived Interferon-Gamma Promotes Hepatocellular Carcinoma Through the Epithelial Cell Adhesion Molecule–Epithelial-to-Mesenchymal Transition Axis in Hepatitis B Virus Transgenic Mice** *Hepatology* **69**:1735–1750

Cheruiyot A *et al.* (2021) **Nonsense-Mediated RNA Decay Is a Unique Vulnerability of Cancer Cells Harboring SF3B1 or U2AF1 Mutations** *Cancer Research* **81**:4499–4513

Falk S, Finogenova K, Melko M, Benda C, Lykke-Andersen S, Jensen TH, Conti E (2016) **Structure of the RBM7–ZCCHC8 core of the NEXT complex reveals connections to splicing factors** *Nature Communications* **7**

Feng P-C, Ke X-F, Kuang H-L, Pan L-L, Ye Q, Wu J-B (2019) **RETRACTED ARTICLE: BMP2 secretion from hepatocellular carcinoma cell HepG2 enhances angiogenesis and tumor growth in endothelial cells via activation of the MAPK/p38 signaling pathway** *Stem Cell Research & Therapy* **10**

Fousek K, Horn LA, Palena C (2021) **Interleukin-8: A chemokine at the intersection of cancer plasticity, angiogenesis, and immune suppression** *Pharmacology & Therapeutics* **219**

Fukushima K *et al.* (2020) **Dysregulated Expression of the Nuclear Exosome Targeting Complex Component Rbm7 in Nonhematopoietic Cells Licenses the Development of Fibrosis** *Immunity* **52**:542–556

Gopinathan G *et al.* (2015) **Interleukin-6 Stimulates Defective Angiogenesis** *Cancer Research* **75**:3098–3107

Harbour JW, Roberson EDO, Anbunathan H, Onken MD, Worley LA, Bowcock AM (2013) **Recurrent mutations at codon 625 of the splicing factor SF3B1 in uveal melanoma** *Nature Genetics* **45**:133–135

Hay N (2011) **Akt isoforms and glucose homeostasis – the leptin connection** *Trends in Endocrinology & Metabolism* **22**:66–73

Inoue D *et al.* (2019) **Spliceosomal disruption of the non-canonical BAF complex in cancer** *Nature* **574**:432–436

Li Z, Guo Q, Zhang J, Fu Z, Wang Y, Wang T, Tang J (2021) **The RNA-Binding Motif Protein Family in Cancer: Friend or Foe?** *Frontiers in Oncology* **11**

Liang Y, Zhang H, Song X, Yang Q (2020) **Metastatic heterogeneity of breast cancer: Molecular mechanism and potential therapeutic targets** *Seminars in Cancer Biology* **60**:14–27

Lubas M, Andersen Peter R, Schein A, Dziembowski A, Kudla G, Jensen Torben H (2015) **The Human Nuclear Exosome Targeting Complex Is Loaded onto Newly Synthesized RNA to Direct Early Ribonucleolysis** *Cell Reports* **10**:178–192

Marimuthu S *et al.* (2022) **MUC16 Promotes Liver Metastasis of Pancreatic Ductal Adenocarcinoma by Upregulating NRP2-Associated Cell Adhesion** *Molecular Cancer Research* **20**:1208–1221

Medeiros B, Allan AL (2019) **Molecular Mechanisms of Breast Cancer Metastasis to the Lung: Clinical and Experimental Perspectives** *International Journal of Molecular Sciences*

Meola N *et al.* (2016) **Identification of a Nuclear Exosome Decay Pathway for Processed Transcripts** *Molecular Cell* **64**:520–533

Montes M, Sanford BL, Comiskey DF, Chandler DS (2019) **RNA Splicing and Disease: Animal Models to Therapies** *Trends in Genetics* **35**:68–87

Pascal Preker JN, Kammler Susanne, Lykke-Andersen Søren, Christensen Marianne S., Mapendano Christophe K., Schierup Mikkel H., Jensen Torben Heick (2008) **RNA Exosome Depletion Reveals Transcription Upstream of Active Human Promoters** *SCIENCE* **322**

Rabbani SA, Mazar AP (2007) **Evaluating distant metastases in breast cancer: from biology to outcomes** *Cancer and Metastasis Reviews* **26**:663–674

Raymond A, Ensslin MA, Shur BD (2009) **SED1/MFG-E8: A Bi-Motif protein that orchestrates diverse cellular interactions** *Journal of Cellular Biochemistry* **106**:957–966

Shi Y (2017) **Mechanistic insights into precursor messenger RNA splicing by the spliceosome** *Nature Reviews Molecular Cell Biology* **18**:655–670

Siegel RL, Miller KD, Jemal A (2019) **Cancer statistics, 2019** *CA: A Cancer Journal for Clinicians* **69**:7–34

Singh RK, Cooper TA (2012) **Pre-mRNA splicing in disease and therapeutics** *Trends in Molecular Medicine* **18**:472–482

Smith MA *et al.* (2019) **U2AF1 mutations induce oncogenic IRAK4 isoforms and activate innate immune pathways in myeloid malignancies** *Nature Cell Biology* **21**:640–650

Tang Y, Horikawa I, Ajiro M, Robles AI, Fujita K, Mondal AM, Stauffer JK, Zheng ZM, Harris CC (2012) **Downregulation of splicing factor SRSF3 induces p53β, an alternatively spliced isoform of p53 that promotes cellular senescence** *Oncogene* **32**:2792–2798

Valastyan S, Weinberg Robert A (2011) **Tumor Metastasis: Molecular Insights and Evolving Paradigms** *Cell* **147**:275–292

Warzecha CC, Sato TK, Nabet B, Hogenesch JB, Carstens RP (2009) **ESRP1 and ESRP2 Are Epithelial Cell-Type-Specific Regulators of FGFR2 Splicing** *Molecular Cell* **33**:591–601

Wilkinson ME, Charenton C, Nagai K (2020) **RNA Splicing by the Spliceosome** *Annual Review of Biochemistry* **89**:359–388

Wright CJ, Smith CWJ, Jiggins CD (2022) **Alternative splicing as a source of phenotypic diversity** *Nature Reviews Genetics* **23**:697–710

Xi P-W, Zhang X, Zhu L, Dai X-Y, Cheng L, Hu Y, Shi L, Wei J-F, Ding Q (2020) **Oncogenic action of the exosome cofactor RBM7 by stabilization of CDK1 mRNA in breast cancer** *npj Breast Cancer* **6**

Yang C *et al.* (2011) **The Integrin αvβ3-5 Ligand MFG-E8 Is a p63/p73 Target Gene in Triple-Negative Breast Cancers but Exhibits Suppressive Functions in ER+ and erbB2+ Breast Cancers** *Cancer Research* **71**:937–945

Zeng L *et al.* (2013) **Vascular Endothelial Cell Growth–Activated XBP1 Splicing in Endothelial Cells Is Crucial for Angiogenesis** *Circulation* **127**:1712–1722

Zhou X *et al.* (2019) **Splicing factor SRSF1 promotes gliomagenesis via oncogenic splice-switching of MYO1B** *Journal of Clinical Investigation* **129**:676–693

Zhou Z, Gong Q, Lin Z, Wang Y, Li M, Wang L, Ding H, Li P (2020) **Emerging Roles of SRSF3 as a Therapeutic Target for Cancer** *Frontiers in Oncology* **10**

Article and author information

Fang Huang
Sino-US Research Center for Cancer Translational Medicine of the Second Affiliated Hospital of Dalian Medical University & Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116023, China

Zhenwei Dai
Sino-US Research Center for Cancer Translational Medicine of the Second Affiliated Hospital of Dalian Medical University & Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116023, China

Jinmiao Yu
Sino-US Research Center for Cancer Translational Medicine of the Second Affiliated Hospital of Dalian Medical University & Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116023, China

Kainan Wang
Department of Oncology & Sino-US Research Center for Cancer Translational Medicine, the Second Affiliated Hospital, Dalian Medical University, Dalian 116023, China

Chaoqun Chen
Sino-US Research Center for Cancer Translational Medicine of the Second Affiliated Hospital of Dalian Medical University & Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116023, China

Dan Chen
Department of Pathology, the First Affiliated Hospital of Dalian Medical University, Dalian Medical University, Dalian 116011, China

Jinrui Zhang
Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116044, China

Jinyao Zhao
Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116044, China

Wenjing Zhang
Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116044, China

Yangfan Qi
Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116044, China
For correspondence: yangfanqi@dmu.edu.cn

Yang Wang
Sino-US Research Center for Cancer Translational Medicine of the Second Affiliated Hospital of Dalian Medical University & Institute of Cancer Stem Cell, Dalian Medical University, Dalian 116023, China
For correspondence: yangwang@dmu.edu.cn
ORCID iD: 0000-0001-9385-7393

Copyright

Editors

Reviewing Editor
Yongliang Yang
Dalian University of Technology, Dalian, China

Senior Editor
Caigang Liu
Shengjing Hospital of China Medical University, Shenyang, China

Reviewer #1 (Public Review):

Summary:
Fang Huang et al found that RBM7 deficiency promotes metastasis by coordinating MFGE8

splicing switch and NF-kB pathway in breast cancer by utilizing clinical samples as well as cell and tail vein injection models.

Strengths:
This study uncovers a previously uncharacterized role of MFGE8 splicing alteration in breast cancer metastasis, and provides evidence supporting RBM7 function in splicing regulation. These findings facilitate the mechanistic understanding of how splicing dysregulation contributes to metastasis in cancer, a direction that has increasingly drawn attention recently, and provides a potentially new prognostic and therapeutic target for breast cancer.

Weaknesses:
This study can be strengthened in several aspects by additional experiments or at least by further discussions. First, how RBM7 regulates NF-kB, and how it coordinates splicing and canonical function as a component of NEXT complex should be clarified. Second, although the roles of MFGE8 splicing isoforms in cell migration and invasion have been demonstrated in transwell and wound healing assays, it would be more convincing to explore their roles in vivo such as the tail vein injection model. Third, the clinical significance would be considerably improved, if the therapeutic value of targeting MFGE8 splicing could be demonstrated.

https://doi.org/10.7554/eLife.95318.1.sa1

Reviewer #2 (Public Review):

Summary:
In this manuscript, the authors reported the biological role of RBM7 deficiency in promoting metastasis of breast cancer. They further used a combination of genomic and molecular biology approaches to discover a novel role of RBM7 in controlling alternative splicing of many genes in cell migration and invasion, which is responsible for the RBM7 activity in suppressing metastasis. They conducted an in-depth mechanistic study on one of the main targets of RBM7, MFGE8, and established a regulatory pathway between RBM7, MFGE8-L/MFGE8-S splicing switch, and NF-κB signaling cascade. This link between RBM7 and cancer pathology was further supported by analysis of clinical data.

Strengths:
Overall, this is a very comprehensive study with lots of data, and the evidence is consistent and convincing. Their main conclusion was supported by many lines of evidence, and the results in animal models are pretty impressive.

Weaknesses:
However, there are some controls missing, and the data presentation needs to be improved. The writing of the manuscript needs some grammatical improvements because some of the wording might be confusing.

Specific comments:
(1) Figure 2. The figure legend is missing for Figure 2C, which caused many mislabels in the rest of the panels. The labels in the main text are correct, but the authors should check the figure legend more carefully. Also in Figure 2C, it is not clear why the authors choose to examine the expression of this subset of genes. The authors only refer to them as "a series of metastasis-related genes", but it is not clear what criteria they used to select these genes for expression analysis.

(2) Line 218-220. The comparison of PSI changes in different types of AS events is misleading. Because these AS events are regulated in different mechanisms, they cannot draw the conclusion that "the presence of RBM7 may promote the usage of alternative splice sites". For

example, the regulators of SE and IR may even be opposite, and thus they should discuss this in different contexts. If they want to conclude this point, they should specifically discuss the SE and A5SS rather than draw an overall conclusion.

(3) In the section starting at line 243, they first referred to the gene and isoforms as "EFG-E8" or "EFG-E8-L", but later used "EFGE8" and "EFGE8-L". Please be consistent here. In addition, it will be more informative if the authors add a diagram of the difference between two EFGE8 isoforms in terms of protein structure or domain configuration.

(4) Figure 7B and 7C. The figures need quantification of the inclusion of MFGE exon7 (PSI value) in addition to the RT-PCR gel. The difference seems to be small for some patients.

Minor comments:
The writing in many places is a little odd or somewhat confusing, I am listing some examples, but the authors need to polish the whole manuscript more to improve the writing.

(1) Line 169-170, "...followed by profiling high-throughput transcriptome by RNA sequencing", should be "followed by high-throughput transcriptome profiling with RNA sequencing".

(2) Line 170, "displayed a wide of RBM7-regulated genes were enriched...", they should add a "that" after the "displayed" as the sentence is very long.

(3) Line 213, "PSI (percent splicing inclusion)" is not correct, PSI stands for "percent spliced in".

(4) Line 216-217, the sentence is long and fragmented, they should break it into two sentences.

(5) Line 224, the "tethering" should be changed to "recognizing". There is a subtle difference in the mechanistic implication between these two words.

(6) Line 250, should be changed to "..in the ratio of two MFGE8 isoforms".

https://doi.org/10.7554/eLife.95318.1.sa0